Lecture # 7

Thursday
28 March 2024
10-11 AM
3-4 PM

PH102
After Mid Semester Exams

by
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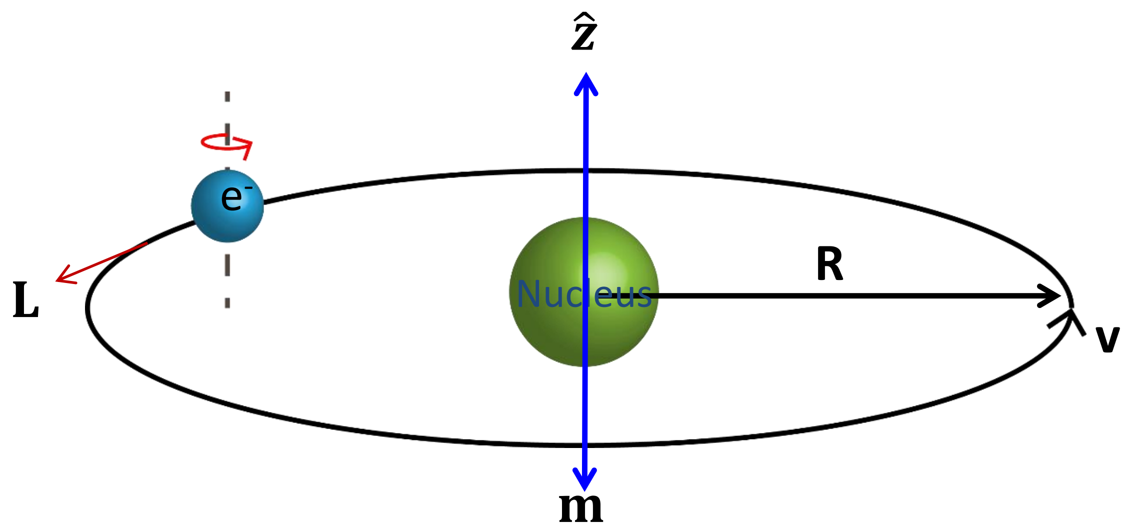
Chapter-7
Electrodynamics

Effect of a Magnetic Field on Atomic Orbits

Electrons not only spin they also revolve around the nucleus--for simplicity, let's assume the orbit is a circle of radius R



$$T = \frac{2\pi R}{v}$$

Although technically this orbital motion does not constitute a steady current in practice the period 'T' is so short that unless you is so short that unless you blink awfully fast, it's going to look like a steady

$$I = \frac{e}{T}$$

$$I = \frac{ev}{2\pi R}$$

Accordingly, the orbital dipole moment is

$$\mathrm{m} = \mathrm{I}\pi R^2$$

$$\mathrm{m} = \frac{ev}{2\pi R}\pi R^2$$

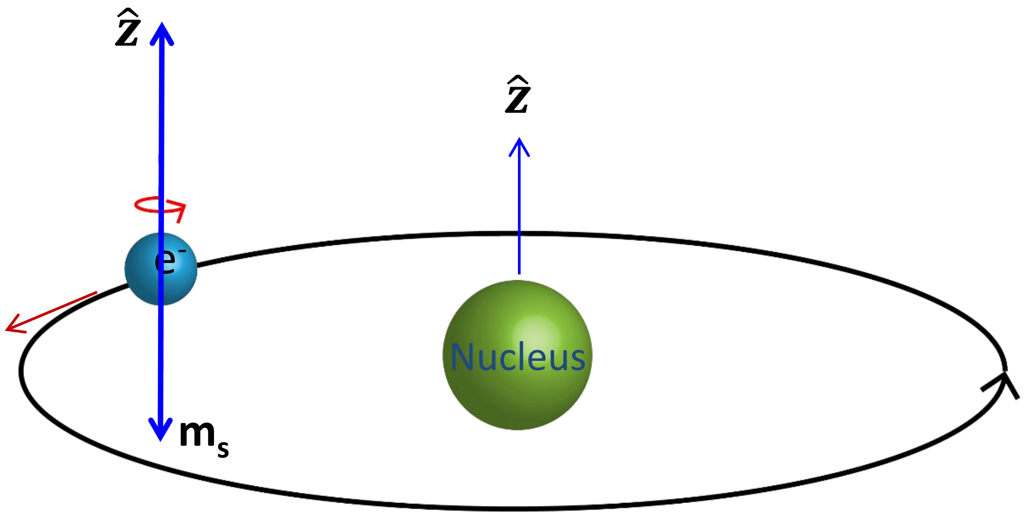
$$\mathrm{m} = \frac{e\ v\ \ R}{2}$$

$$\boxed{\mathbf{m} = -\frac{\mathbf{e\ v\ \pi\ R}}{\mathbf{2}}\ \hat{\boldsymbol{z}}}$$

Here, minus sign accounts for the negative charge of the electron

In the presence of Magnetic field it will experience a Torque

But it's a lot harder to tilt the entire orbit than it is the spin, so the orbital contribution to Paramagnetism is small.



Instead of tilting the entire orbit, the electron speeds up or slows down depending on the orientation of magnetic field **B**.

Here, the centripetal acceleration is ordinarily sustained by electrical forces alone.

Electrical Coulomb force $\longrightarrow$ $$\frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2}=\frac{m_e v^2}{r}$$ Eq.1 $\longleftarrow$ Centripetal force

In the presence of a magnetic field there is an additional magnetic force

$$\overrightarrow{F_m}=-e\left(\overrightarrow{v'}\times\vec{B}\right)$$ $\longleftarrow$ Magnetic force

For the sake of argument, let's say that B is perpendicular to the plane of the orbit.

$$F_m=ev'B$$

The total force

$$\frac{1}{4\pi \in_0}\frac{e^2}{r^2} + ev'B = \frac{m_e v'^2}{r}$$

Eq.2

Under these conditions, the new speed v′ is greater than v

After subtracting Eq.2 from Eq.1

$$\frac{1}{4\pi \in_0}\frac{e^2}{r^2} = \frac{m_e v^2}{r}$$

$$ev'B = \frac{m_e}{r}(v'^2 - v^2)$$

$$ev'B = \frac{m_e}{r}(v' - v)(v' + v)$$

Here, $\Delta v = v' - v$

$$erB = \frac{m_e}{v'}(\Delta v)(v' + v)$$

$$erB = m_e(\Delta v)\left(1 + \frac{v}{v'}\right)$$

$$\frac{v}{v'} \rightarrow 1$$

If Δv is very small

$$erB = 2m_e(\Delta v)$$

$$\boxed{\Delta v = \frac{erB}{2m_e}}$$

When B is turned on, then, the electron speeds up.

A change in orbital speed means a change in the dipole moment

$$\Delta m = -\frac{1}{2}e\,\Delta v\,R\,\hat{z}$$

since

$$m = -\frac{e\,v\,\pi\,R}{2}\,\hat{z}$$

$$\Delta m = -\frac{e^2R^2}{4me}B$$

Note that the change in **m** is opposite to the direction of B

In diamagnetic materials, the magnetic susceptibility can be accurately predicted by Langevin's classical theory of electromagnetism as follows :

$$\boxed{\chi_m = \frac{-\mu_0 n Z e^2 < r^2 >}{6 m_0}}$$

Where :

χ_m : frequently a susceptibility is defined referred to unit mass or to a mole of the substance , which means The molar susceptibility .

μ_0 : the magnetic permeability of a vacuum in H.

Z : the atomic number of the atom .

n : the atomic density in m^{-3}

e : the elementary charge in C .

$< r^2 >$: the root mean square of the square of the at

The magnetic moment of an atom with Z electrons

$$\mu_{atom} = -\frac{Ze^2 \delta B}{6m}\langle r^2 \rangle$$

Diamagnetic susceptibility per unit volume

$$\chi = \frac{\mu_o N \mu_{atom}}{\delta B}$$

Number of atoms per unit volume

$$\boxed{\chi = -\frac{\mu_o N Z e^2}{6m}\langle r^2 \rangle}$$

Classical Langevin equation for diamagnetism

10.4.3 Discussion

On the basis of quantum orbital theory, the mean value of r^2 for an orbit about an effective nuclear charge Z is given by

$$\overline{r^2} = a_0^2 \cdot \frac{n^2}{Z^2}\left(\frac{5}{2}n^2 - \frac{3}{2}k^2\right) \tag{10.44}$$

where $a_0(= 0.528 \times 10^{-10}\ \text{m})$ is the radius of the innermost orbit in hydrogen atom, n is the radial quantum number and k, the azimuthal quantum number.

This gives for the contribution of the orbit to the gram atomic diamagnetic susceptibility

$$\chi_A = -2.83 \times 10^{10}\overline{r^2}$$

$\Rightarrow$

$$\chi_A = -0.785 \times 10^{-6}\left[\frac{n^2}{Z^2}\left(\frac{5}{2}n^2 - \frac{3}{2}k^2\right)\right] \tag{10.45}$$

If the effective nuclear charge is calculated from this expression for helium with two electrons from its observed susceptibility ($\chi_A = -1.9 \times 10^{-6}$), it comes out to be 0.93 which is too small.

To make the expression for susceptibility more accurate, van Vleck and Pauling modified the above expression as

$$\chi_A = -0.785 \times 10^{-6}\left[\frac{n^2}{Z^2}\left(\frac{5}{2}n^2 - \frac{3l(l+1)-1}{2}\right)\right] \tag{10.46}$$

where l, the orbital quantum number, is equal to $(k-1)$.

Pauling's calculations necessarily involve a number of approximations. For spherically symmetric atoms, Hartree has devised a method with which the charge distribution satisfying the Schrödinger equation may be worked out more precisely. He has given tables and curves for a number of ions and atoms showing the charge per unit radial distance in a spherical shell of unit thickness.

If $\frac{dN}{dr}$ be the charge in electron unit per unit radial distance, then the number of electrons in the ions is equal to $\int_0^\infty \left(\frac{dN}{dr}\right) dr$.

For the diamagnetic susceptibility, he obtained

$$\chi_A = -0.785 \times 10^{-6}\left[\frac{n^2}{Z^2}\left(\frac{5}{2}n^2 - \frac{3}{2}k^2\right)\right]$$

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$$\chi_A = -2.83 \times 10^{10} \int_0^\infty \left(\frac{dN}{dr}\right) dr$$

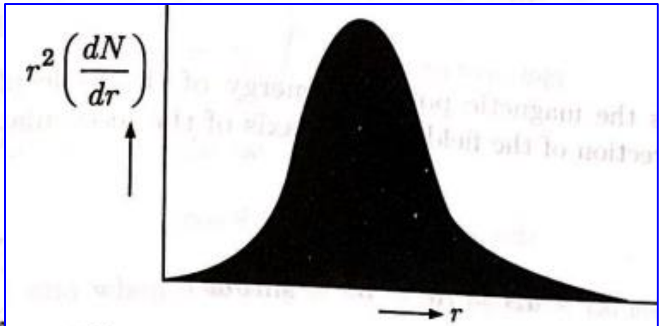


Figure 10.2: Variation of $r^2(dN/dr)$ with r.

This integral can be evaluated graphically (Figure 10.2). Using atomic units for distances, the susceptibility is given by

$$\chi_A = -0.785 \times 10^{-6} \times \text{ area under the shaded portion of the curve}$$

For helium this gives

$$\chi_A = -1.89 \times 10^{-6}$$

a value which agrees remarkably closely with the observed one (-1.9×10^{-6}).

- Ordinarily, the electron orbits are randomly oriented, and the orbital dipole moments cancel out. But in the presence of a magnetic field, each atom picks up a little "extra" dipole moment, and these increments are all antiparallel to the field.

- This is the mechanism responsible for diamagnetism. It is a universal phenomenon, affecting all atoms. However, it is typically much weaker than paramagnetism, and is therefore observed mainly in atoms with even numbers of electrons. where paramagnetism is usually absent.

- The diamagnetism is a quantum phenomenon not the classical.

- A complete understanding of these atomic mechanisms requires application of quantum mechanics beyond the scope of this PH-102.

Example 4: Among the following materials, which would you expect to be paramagnetic and which diamagnetic?

- ❑ Aluminum
- ❑ Copper
- ❑ Nitrogen (N_2)
- ❑ Salt (NaCl)
- ❑ Sodium (Na)
- ❑ Copper chloride ($CuCl_2$)
- ❑ Carbon
- ❑ Lead
- ❑ Sulfur
- ❑ Water (H_2O)

Solution: Usually, the materials having odd number of electrons exhibits Paramagnetic nature and even number of electrons materials exhibits Diamagnetism.

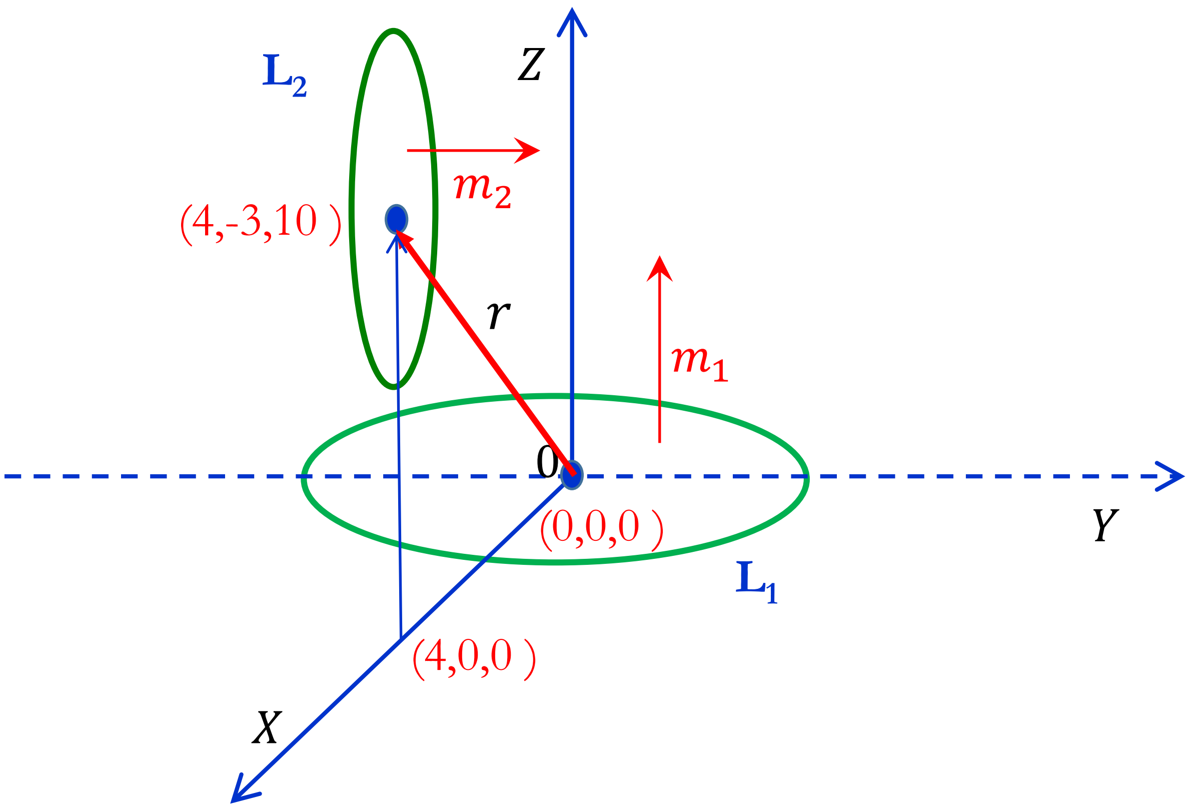
S.N.	Name of Materials	Number of Electrons	Magnetic Nature
1	Aluminum	13	Paramagnetic
2	Copper*	29	Paramagnetic
3	Copper Chloride	45	Paramagnetic
4	Carbon	6	Diamagnetic
5	Lead	82	Diamagnetic
6	Nitrogen (N_2)	14	Diamagnetic
7	Sodium Chloride	28	Diamagnetic
8	Sodium	11	Paramagnetic
9	Sulfur	16	Diamagnetic
10	Water	10	Diamagnetic

*Note: Copper is slightly diamagnetic; otherwise they're all what we have predicted.

Few Examples

Problem

A small current loop L_1 with magnetic moment $53\hat{z}\ A.m^2$ is located at the origin while another small loop current L_2 with magnetic moment $3\ \hat{y}\ A.m^2$ is located at (4,-3,10) . Determine the torque on L_2.



Solution 1:

The torque T_2 on the loop L_2 is due to the field B_1 which is produced by loop L_1 is ,

$$T_2 = \underset{\boxed{?}}{m_2} \times \underset{\boxed{?}}{B_1}$$

Since m_1 is magnetic moment for loop L_1 is along $\hat{z}$.

Magnetic flux density due to circular current loop L_1 is

$$\boxed{\overrightarrow{B_1} = \frac{\mu_0 m}{4\pi r^3}\left(2cos\theta\hat{r} + sin\theta\widehat{\Theta}\right)}$$

Given $m_2 = 3\,\hat{y} = 3(\sin\theta\sin\phi\hat{r} + \cos\theta\sin\phi\widehat{\Theta} + \cos\phi\hat{\phi})$

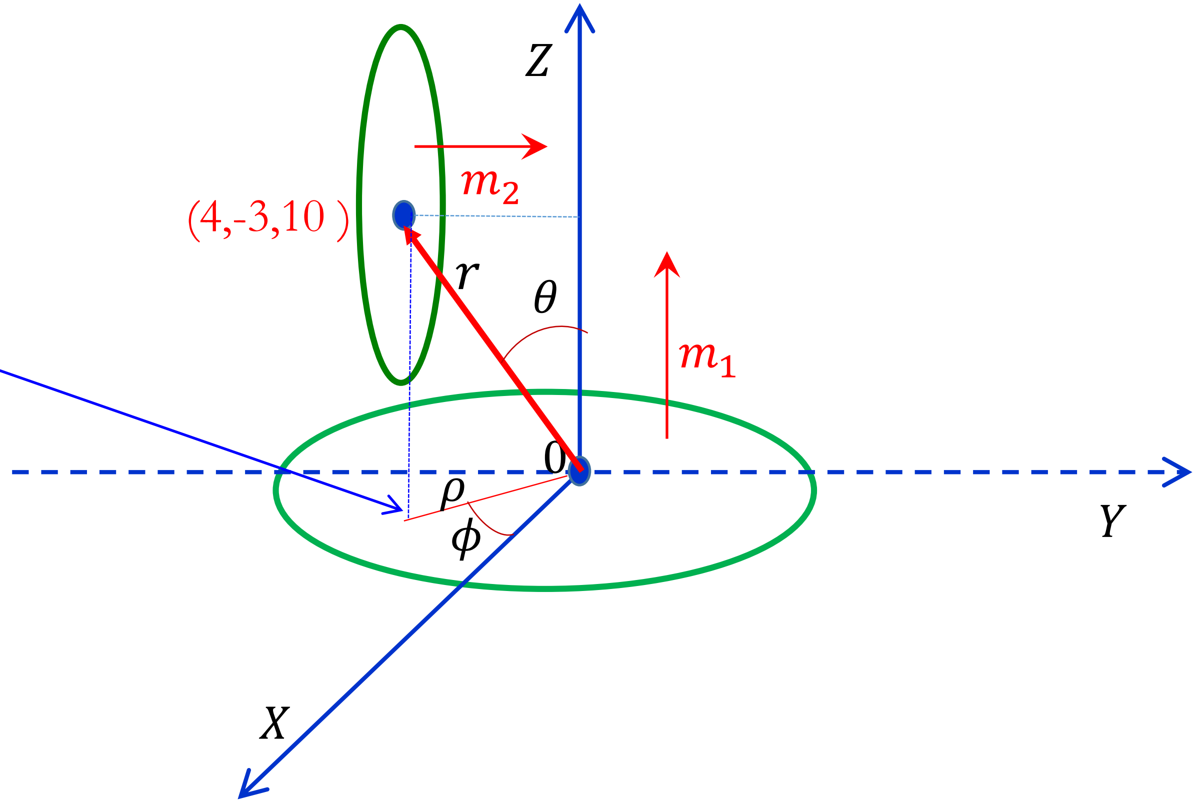
(in spherecal coordinate)

At (4,-3,10) distance between two circular loop is

$$r = \sqrt{4^2 + (-3)^2 + 10^2} = 5\sqrt{5}$$

From figure

$$\rho = \sqrt{4^2 + (-3)^2}$$



$$tan\theta = \frac{\rho}{z} = \frac{5}{10} = \frac{1}{2}$$

$$\longrightarrow sin\theta = \frac{\rho}{r} = \frac{1}{\sqrt{5}}$$

$$\longrightarrow cos\theta = \frac{z}{r} = \frac{2}{\sqrt{5}}$$

$$tan\phi = \frac{y}{x} = \frac{-3}{4}$$

$$\longrightarrow sin\phi = \frac{-3}{5}$$

$$\longrightarrow cos\phi = \frac{4}{5}$$

$$\overrightarrow{B_1} = \frac{\mu_0 m}{4\pi r^3}\left(2cos\theta\hat{r} + sin\theta\widehat{\Theta}\right)$$

Substituting all the value we get magnetic field ,

$$B = \frac{4\pi\times10^{-7}\times5}{4\pi625\sqrt{5}}\left(\frac{4}{\sqrt{5}}\hat{r} + \frac{1}{\sqrt{5}}\widehat{\Theta}\right)$$

$$B = \frac{10^{-7}}{625}\left(4\hat{r} + \widehat{\Theta}\right)$$

$$m_2 = 3\left(-\frac{3}{5\sqrt{5}}\hat{r} - \frac{6}{5\sqrt{5}}\hat{\theta} + \frac{4}{5}\hat{\phi}\right)$$

$$T = \frac{10^{-7}(3)}{625(5\sqrt{5})}\left(-3\hat{r} - 6\widehat{\Theta} + 4\sqrt{5}\hat{\phi}\right) \times \left(4\hat{r} + \hat{\phi}\right)$$

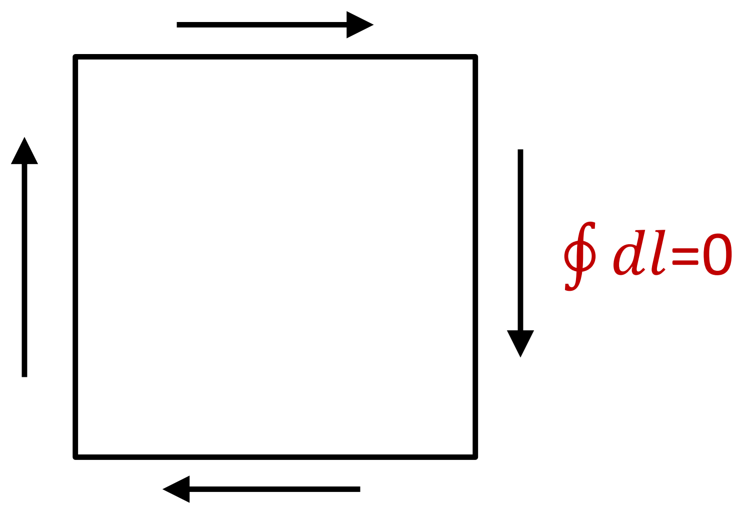
$$T = -0.258\hat{r} + 1.665\widehat{\widehat{\Theta}} + 1.03\hat{\phi}\ \mathrm{nN.m}$$

Forces in the presence of Magnetic fields

In the presence of uniform External magnetic field, the net force on a current loop is zero

$$F = I \oint (\boldsymbol{dl} \times \boldsymbol{B})$$

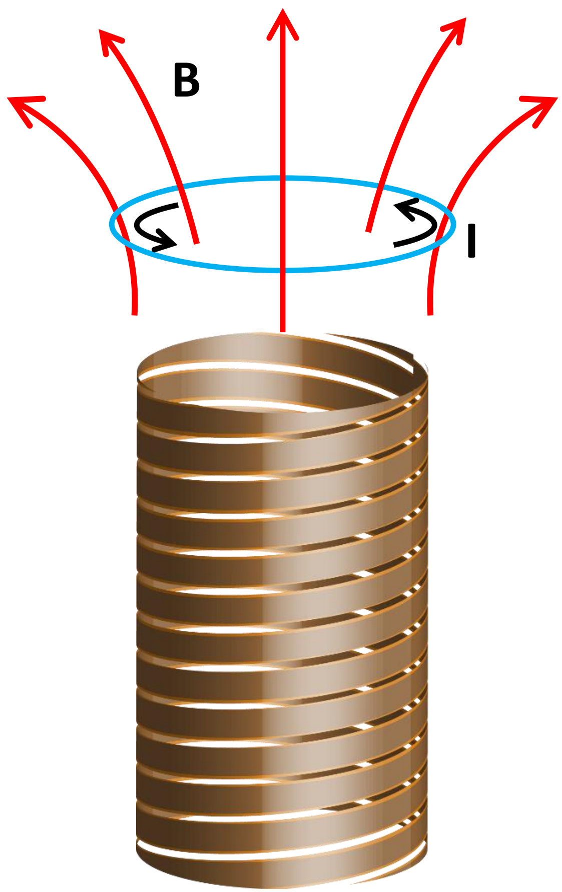
Since 'B' is a constant/uniform, this comes outside from the integral, and the net displacement $\oint dl$ around a closed loop vanishes.



Hence $F = 0$

However, For the non-uniform Magnetic fields this Force does not vanishes

Example: Suppose a circular wire of radius ‘R’, carrying a current ‘I’, is suspended above a short solenoid in the "fringing" region as shown in the below figure.



Here **B** has a radial component, and there is a net downward force on the loop (Fig. 2):

$$F = I\ (\boldsymbol{l} \times \boldsymbol{B})$$

$$l = 2\pi R$$

$$\boxed{\mathrm{F} = 2\pi \mathrm{IRB} \cos \theta}$$

$$\boxed{\mathrm{F} = -2\pi \mathrm{IRB} \cos \theta\ \hat{z}}$$

Net force on loop is downward along negative z-direction (see page 257 of Griffiths, Eq: 6.2).

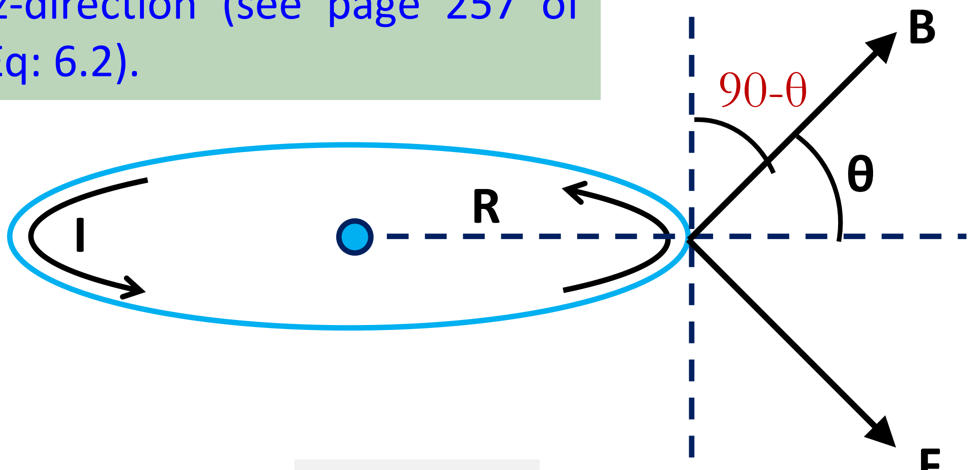


Figure 2

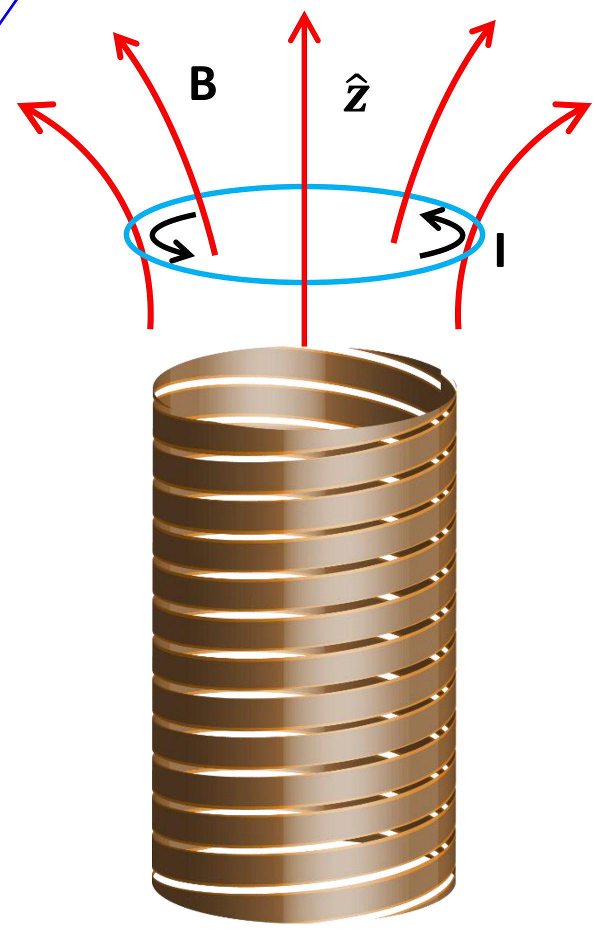
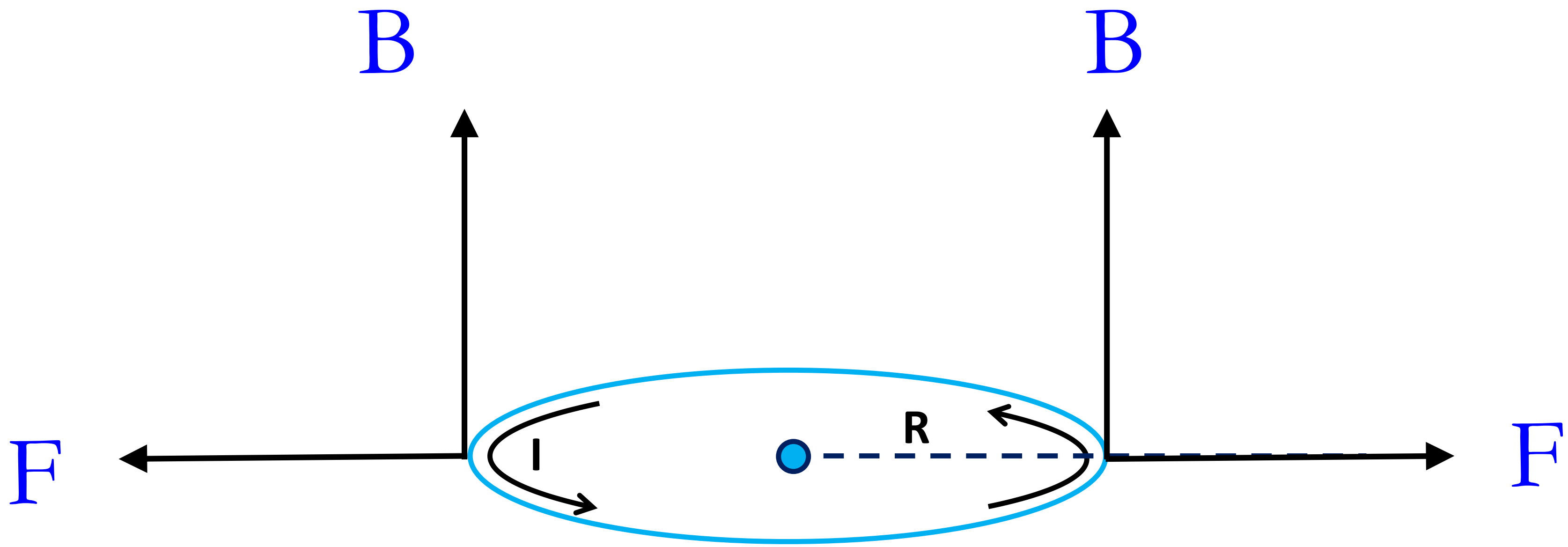
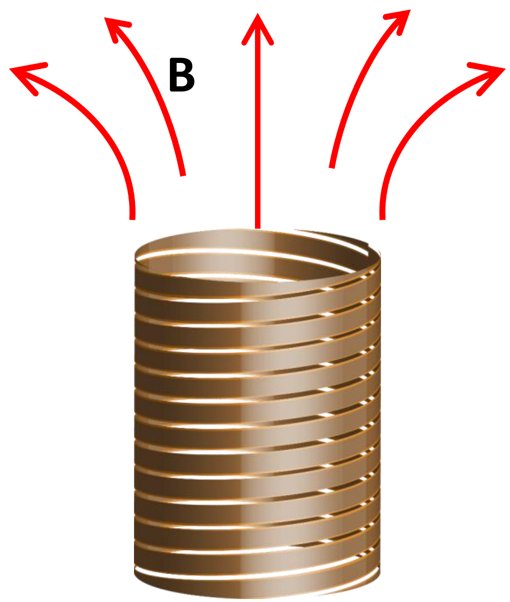


Figure 1



No net force experienced by the current loop in the presence of Uniform external magnetic field (situation when solenoid is very far away from the loop). Magnetic field 'B' will be along the Z axis.



For an infinitesimal loop, with dipole moment **m**, in an external magnetic field B, the force is

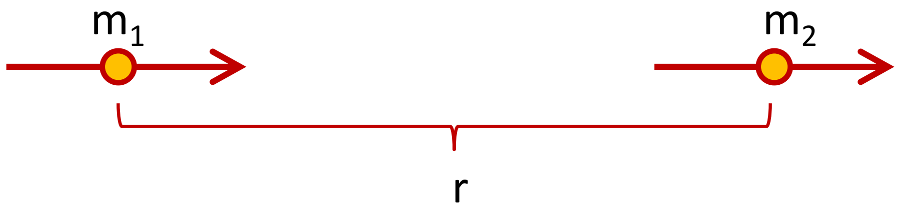
$$\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B})$$

Once again the magnetic formula is identical to its electrical "twin," provided we agree to write the latter in the form

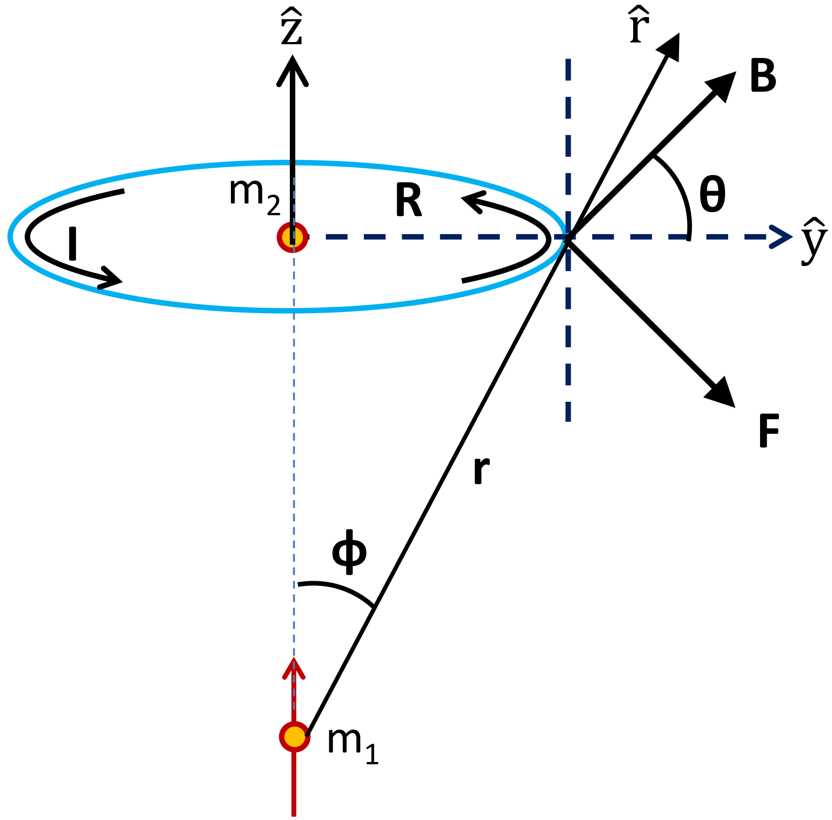
$$\mathbf{F} = \nabla(\mathbf{p} \cdot \mathbf{E})$$

Example : Find the force of attraction between two magnetic dipoles, m_1 and m_2, oriented as shown in below Figure, a distance r apart.

Method-I

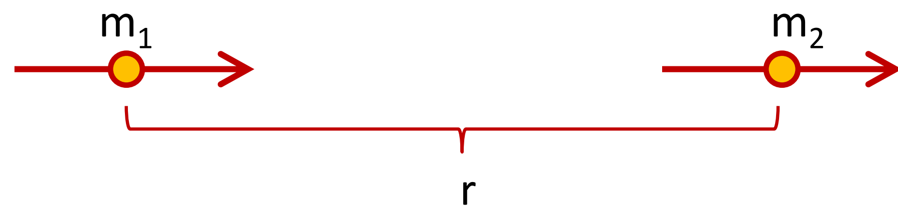


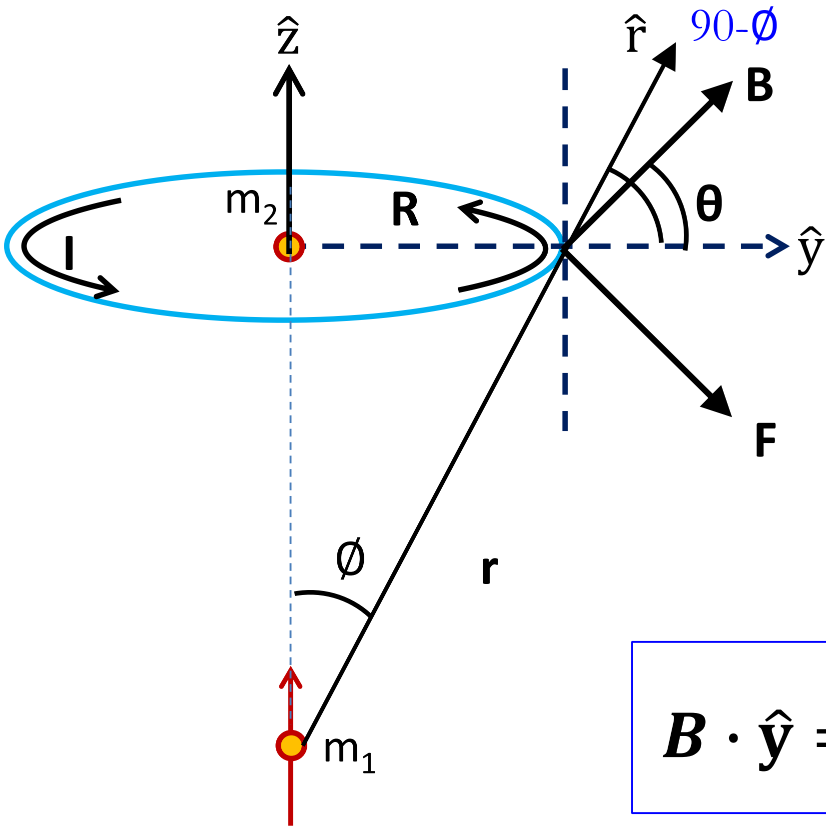
Method-I



The magnetic field produced by m_1 at m_2 (r) is non uniform

Method-I





We know that

$$F = 2\pi I R B \cos\theta$$

$$\boldsymbol{B} = \frac{\mu_0}{4\pi}\frac{1}{r^3}\left[3(\boldsymbol{m_1}\cdot\hat{r})\hat{r} - \boldsymbol{m_1}\right]$$

$$B\cos\theta = \mathbf{B}\cdot\hat{\mathbf{y}}$$

$$\boldsymbol{B}\cdot\hat{\mathbf{y}} = \frac{\mu_0}{4\pi}\frac{1}{r^3}\left[3(\boldsymbol{m_1}\cdot\hat{r})(\hat{r}\cdot\hat{y}) - (\boldsymbol{m_1}\cdot\hat{y})\right]$$

But $\boldsymbol{m_1}\cdot\hat{y} = 0$ Since m_1 is in the z-direction

$$\boldsymbol{m_1}\cdot\hat{r} = m_1 \cos\emptyset \qquad \hat{r}\cdot\hat{y} = \sin\emptyset$$

$$\mathrm{B}\cos\theta = \mathbf{B}\cdot\hat{\mathbf{y}}$$

$$\mathbf{B}\cdot\hat{\mathbf{y}} = \frac{\mu_0}{4\pi}\frac{1}{r^3}[3(\boldsymbol{m}_{\mathbf{1}}\cdot\hat{r})(\hat{r}\cdot\hat{y}) - (\boldsymbol{m}_{\mathbf{1}}\cdot\hat{y})]$$

$$\boldsymbol{m}_{\mathbf{1}}\cdot\hat{r} = m_1 \ \cos\emptyset$$

$$\mathbf{m}_{\mathbf{1}}\cdot\hat{\mathrm{y}} = 0$$

$$\hat{r}\cdot\hat{y} = \sin\emptyset$$

$$\mathrm{B}\cos\theta \ = \frac{\mu_0}{4\pi}\frac{1}{r^3}[3(m_1\cos\emptyset)(\sin\emptyset)\ ]$$

$$\mathrm{B}\cos\theta = \frac{\mu_0}{4\pi}\frac{1}{r^3}[3(m_1\cos\emptyset)(\sin\emptyset)]$$

$$F = 2\pi IRB\cos\theta$$

$$F = 2\pi IR\frac{\mu_0}{4\pi}\frac{1}{r^3}[3(m_1\cos\emptyset)(\sin\emptyset)]$$

$$\sin\emptyset = \frac{R}{r} \qquad \cos\emptyset = \frac{\sqrt{r^2 - R^2}}{r}$$

$$F = 2\pi IR\frac{\mu_0}{4\pi}\frac{1}{r^3}\left[3\left(m_1\frac{\sqrt{r^2 - R^2}}{r}\right)\right]\frac{R}{r}$$

But $\rightarrow$ $IR^2\pi = m_2$

$$F = \frac{3\mu_0}{2\pi}\frac{m_1 m_2}{r^5}\sqrt{r^2 - R^2}$$

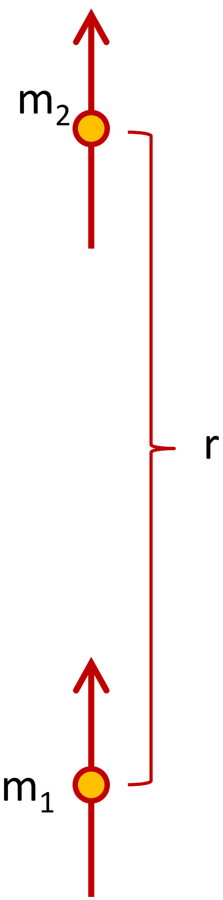
$$F = \frac{3\mu_0}{2\pi}\frac{m_1 m_2}{r^5}\sqrt{r^2\left(1-\left(\frac{R}{r}\right)^2\right)}$$

While for a dipole

$$R \ll r$$

$$F = \frac{3\mu_0}{2\pi}\frac{m_1 m_2}{r^4}$$

Net force on loop is downward along negative z-direction (see page 257 of Griffiths, Eq: 6.2) and See Slides of this lecture.



Method-II

$$F = \nabla\,(\boldsymbol{m_2}\cdot\boldsymbol{B})$$

$$F = \,(\boldsymbol{m_2}\cdot\nabla)\,\boldsymbol{B}$$

$$F = \left(m_2\cdot\frac{d}{dz}\right)\left[\frac{\mu_0}{4\pi}\frac{1}{z^3}[3(\boldsymbol{m_1}\cdot\hat{z})\hat{z} - \boldsymbol{m_1}]\right]$$

$$\boxed{F = \left(m_2 \cdot \frac{d}{dz}\right)\left[\frac{\mu_0}{4\pi}\frac{1}{z^3}[3(\boldsymbol{m_1} \cdot \hat{z})\hat{z} - \boldsymbol{m_1}]\right]}$$

Here $\rightarrow$ $\mathbf{m_1} = m_1\hat{z}$

$$F = \left(m_2 \cdot \frac{d}{dz}\right)\left[\frac{\mu_0}{4\pi}\frac{2m_1}{z^3}\right]$$

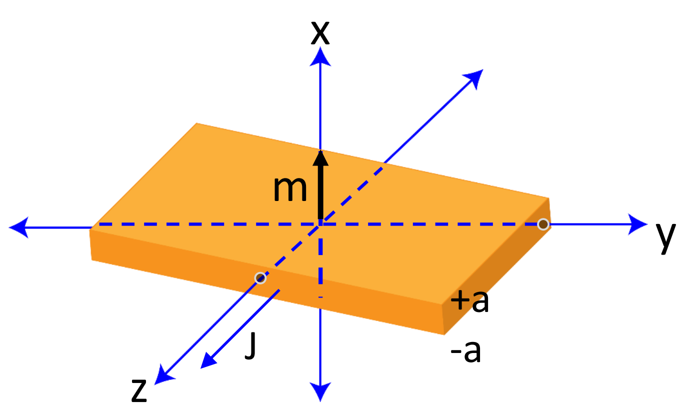
$$\boldsymbol{F} = -\frac{3\mu_0}{2\pi}\frac{m_2 m_1}{z^4}\hat{z}$$

(see page 257 of Griffiths, Eq: 6.2) and See Slide #30 of this lecture.

Net force on loop is downward along negative z-direction.

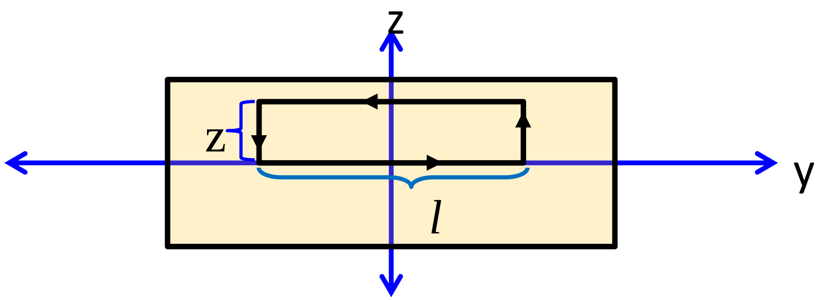
Example : A uniform current density $J = J_0\,\hat{z}$ fills a slab straddling the yz plane, from x = -a to x = +a. A magnetic dipole $\boldsymbol{m} = m_0\,\hat{x}$ is situated at the origin.

a. Find the force on the dipole using the equation $F = \nabla\,(\boldsymbol{m}\cdot\boldsymbol{B})$.
b. Do the same for a dipole pointing in the y direction: $\boldsymbol{m} = m_0\,\hat{y}$.
c. In the electrostatic case the expressions F $= \nabla\,(\boldsymbol{p}\cdot\boldsymbol{E})$. and F $= (\boldsymbol{p}\cdot\nabla)\boldsymbol{E}$. are equivalent (prove it), but this is not the case for the magnetic analogs (explain why). As an example, calculate $(\boldsymbol{m}\cdot\nabla)\boldsymbol{B}$ for the configurations in (a) and (b).



Solution: As a general case, first we will calculate the field due to a magnetic slab $J = J_0\,\hat{x}$ lying in the *xy* plane from z = -a to +a (see the figure 5.1)

Finally we will modify this result for the present given condition i.e. $J = J_0\,\hat{z}$



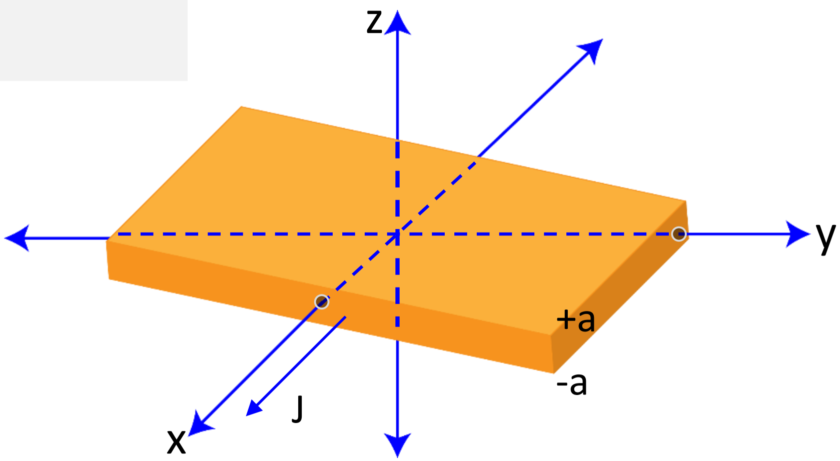


Figure 5.1

Using the amperian loop as shown in this figure

$$\oint \vec{B}\cdot\vec{dl} = Bl = \mu_0 I_{enclosed} \rightarrow Bl = \mu_0 lz\mathrm{J} \rightarrow B = \mu_0 z\mathrm{J}$$

$$I_{enclosed} = l\,z\,\mathrm{J}$$

$$B = -\mu_0 z\mathrm{J}\,\hat{y}$$

For z < a

If z > a $\quad I_{enclosed} = l\, a\, J$

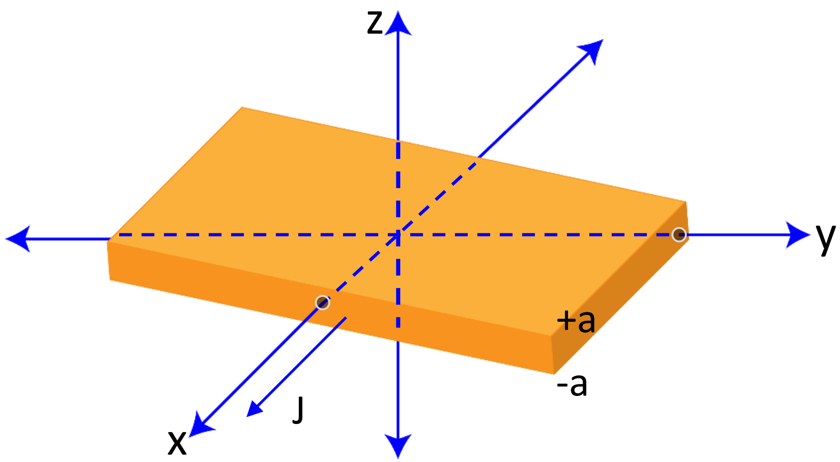
So

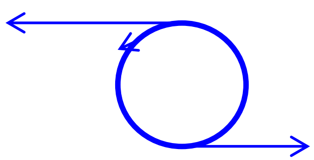
$$B = -\mu_0 J a\, \hat{y}$$

z > +a

$$B = +\mu_0 J a\, \hat{y}$$

z > -a



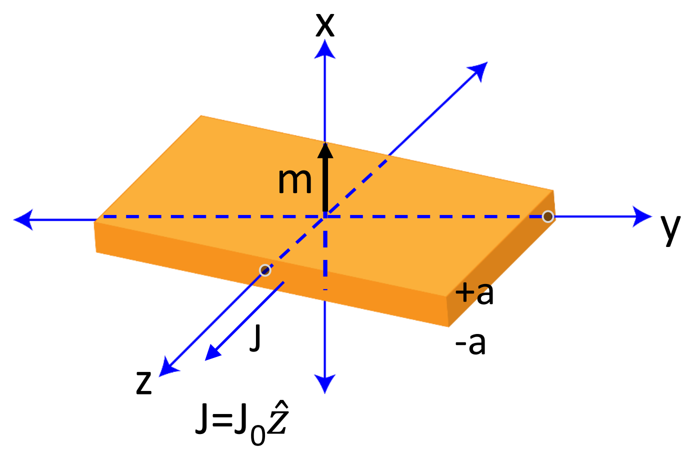


For the present given condition; $J = J_0\,\hat{z}$ with slab straddling the yz plane, from x = -a to x = +a.

Part a

Here the direction of magnetic field remain the same (check using right hand rule)

Everything will remain same in the above equation except the Amperian loop will lie in the xy plane.



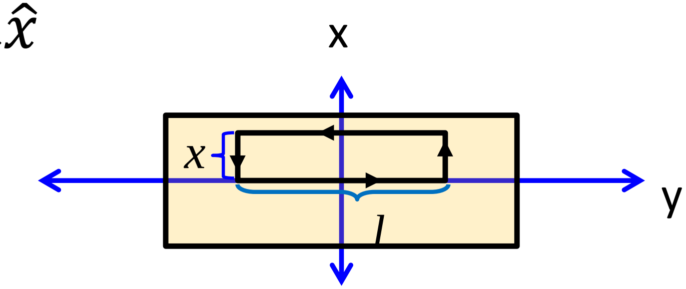
So the magnetic field will be

$$B = -\mu_0 J_0 x\,\hat{y}$$

$\boldsymbol{m}\cdot\boldsymbol{B} = 0$ Because $\boldsymbol{m} = m\hat{x}$

$$F = \nabla\,(\boldsymbol{m}\cdot\boldsymbol{B})$$

$$F = 0$$



Part b

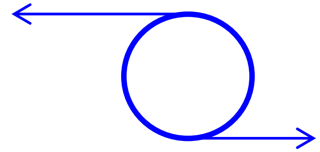
$$B = -\mu_0 J_0 \mathrm{x}\,\hat{y}$$

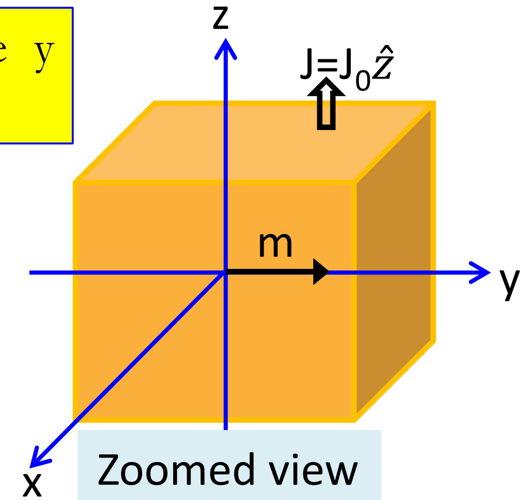
$$m = m_0\,\hat{y}$$

$$\boldsymbol{m}\cdot\boldsymbol{B} = m_0\mu_0 J_0 x$$

$$\boldsymbol{F} = m_0\mu_0 J_0\hat{x}$$

For a dipole pointing in the y direction: $m = m_0\,\hat{y}$





Part c

In Electrostatics

$$\mathrm{F} = \nabla\,(\mathbf{p}\cdot\mathbf{E})$$

$$\nabla\,(\mathbf{p}\cdot\mathbf{E}) = (\vec{\mathrm{p}}\cdot\vec{\nabla})\vec{\mathrm{E}}$$

Using the formula

$$\nabla\,(\boldsymbol{A}\cdot\boldsymbol{B}) = \vec{A}\times(\vec{\nabla}\times\vec{B}) + \vec{B}\times(\vec{\nabla}\times\vec{A}) + (\vec{A}\cdot\vec{\nabla})\vec{B} + (\vec{B}\cdot\vec{\nabla})\vec{A}$$

$$\nabla\,(\boldsymbol{p}\cdot\boldsymbol{E}) = \vec{p}\times\underbrace{(\vec{\nabla}\times\vec{E})}_{\vec{\nabla}\times\vec{E}=0} + \vec{E}\times\underbrace{(\vec{\nabla}\times\vec{p})}_{0} + (\vec{p}\cdot\vec{\nabla})\vec{E} + \underbrace{(\vec{E}\cdot\vec{\nabla})\vec{p}}_{0}$$

Since p does not depend on (x,y,z) the 2nd and 4th term will become zero

$$\boxed{\nabla\,(\mathbf{p}\cdot\mathbf{E}) = (\vec{\mathrm{p}}\cdot\vec{\nabla})\vec{\mathrm{E}}}$$

In Magnetostatics

Proof

$$\mathrm{F} = \nabla\,(\mathbf{m}\cdot\mathbf{B})$$

$$\nabla\,(\boldsymbol{m}\cdot\boldsymbol{B}) = \vec{m}\times(\vec{\nabla}\times\vec{B}) + \vec{B}\times(\underbrace{\vec{\nabla}\times\vec{m}}_{0}) + (\vec{m}\cdot\vec{\nabla})\vec{B} + (\underbrace{\vec{B}\cdot\vec{\nabla}}_{0})\vec{m}$$

However in magnetostatics $\vec{\nabla}\times\vec{B}\neq 0$

Hence

$$\boxed{\nabla\,(\mathbf{m}\cdot\mathbf{B}) \neq (\vec{\mathrm{m}}\cdot\vec{\nabla})\vec{\mathrm{B}}}$$

$$\nabla\,(\boldsymbol{m}\cdot\boldsymbol{B}) = \vec{m}\times(\vec{\nabla}\times\vec{B}) + (\vec{m}\cdot\vec{\nabla})\vec{B}$$

$$\vec{\nabla}\times\vec{B} = \mu_0 J$$

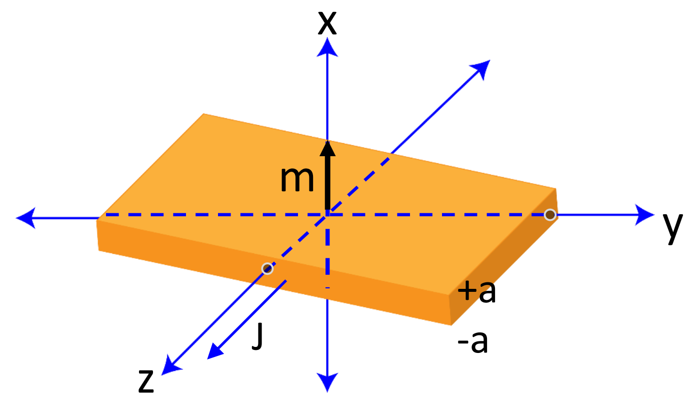
$$\boxed{\nabla\,(\boldsymbol{m}\cdot\boldsymbol{B}) = \mu_0\mathrm{J}(\vec{m}\times\vec{B}) + (\vec{m}\cdot\vec{\nabla})\vec{B}}$$

$(\mathbf{m}\cdot\nabla)\mathbf{B}$ for part (a) and part (c)

For part (a)

$$(\boldsymbol{m}\cdot\nabla)\boldsymbol{B} = m_0\frac{d}{dx}(B)$$

$$B = -\mu_0 J_0 \mathrm{x}\,\hat{y}$$



Here the direction of $\mathbf{m}$ is in $\hat{x}$ **direction**

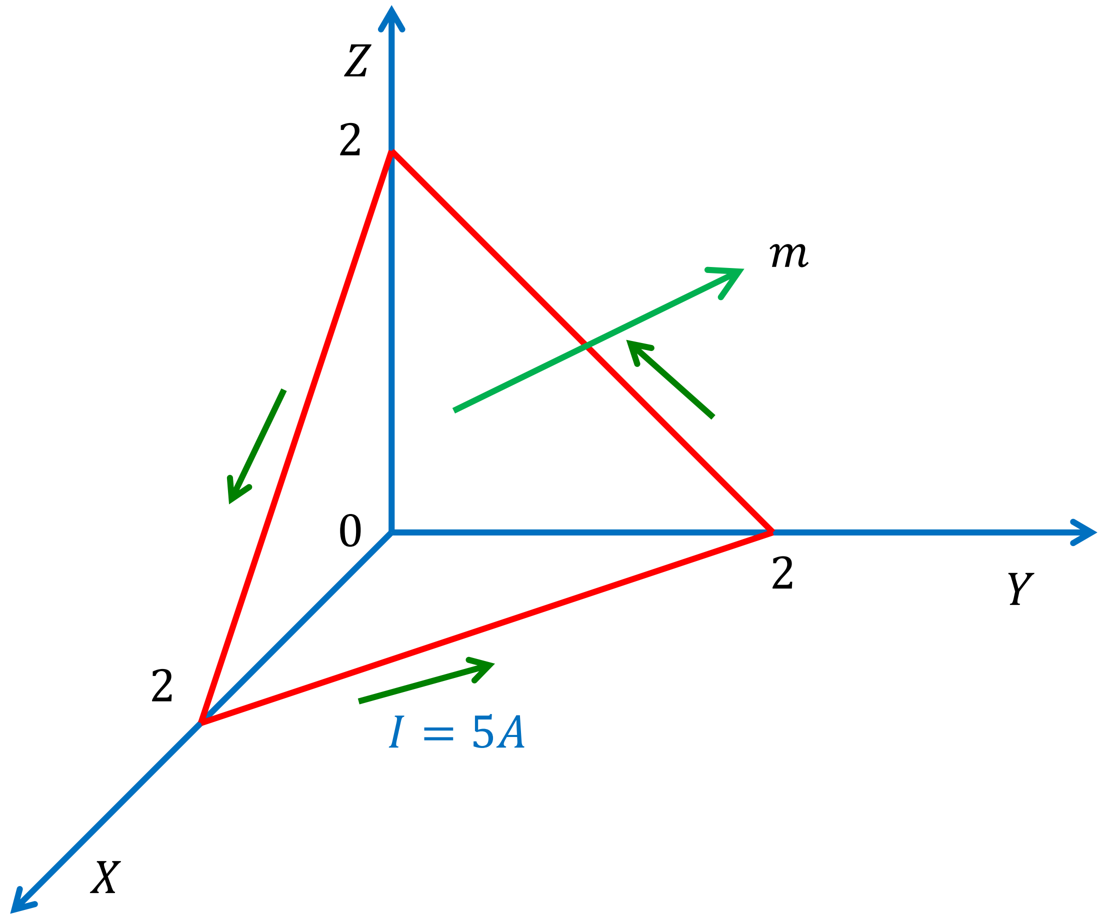
$$(\boldsymbol{m}\cdot\nabla)\boldsymbol{B} = m_0\frac{d}{dx}(B) = -m_0\mu_0 J_0\,\hat{y}$$

For part (b)

$$(\boldsymbol{m}\cdot\nabla)\boldsymbol{B} = m_0\frac{d}{dy}(B)$$

$$(\boldsymbol{m}\cdot\nabla)\boldsymbol{B} = m_0\frac{d}{dy}(B) = 0$$

Determine the magnetic moment of an electric circuit formed by the triangular loop .



The general equation of a plane is given by : $Ax + By + Cz + D = 0$

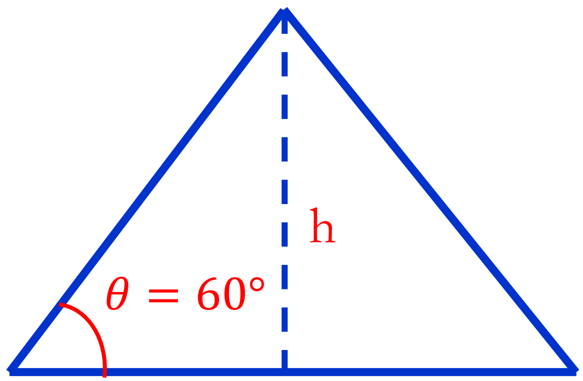
where , $D = -(A^2 + B^2 + C^2)$

Since points $(2,0,0)$, $(0,2,0)$,and $(0,0,2)$ lie on the plane, these points must satisfy the equation of the plane .

Equation of the triangular plane on which the loop lies. : $x + y + z = 2$

Magnetic moment of an electric circuit $m = I\,S\hat{n}$

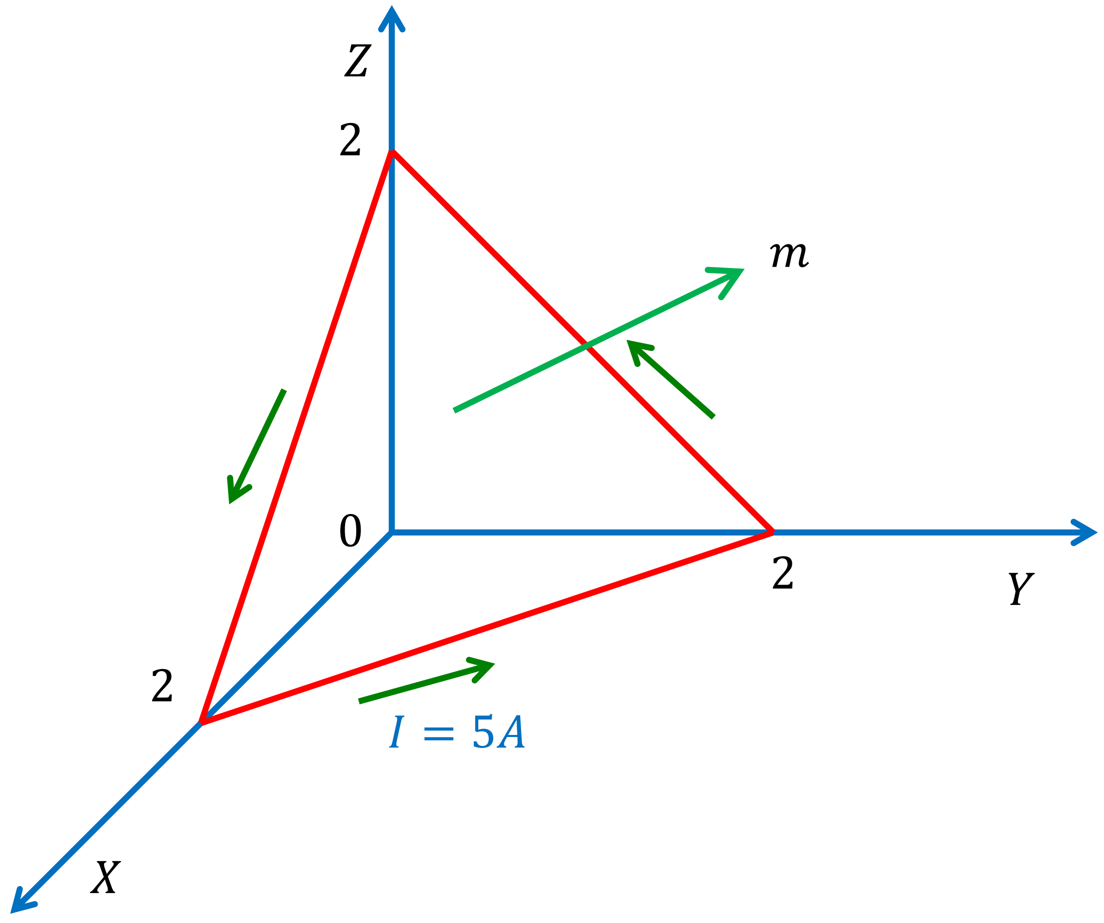
Where , $S =$ loop area $=1/2 \times$ base $\times$ height



From figure , base$=\sqrt{2^2 + 2^2} = 2\sqrt{2}$

height(h)$=\sqrt{2^2 + 2^2}\,sin60° = 2\sqrt{2}\,sin60°$

Determine the magnetic moment of an electric circuit formed by the triangular loop .



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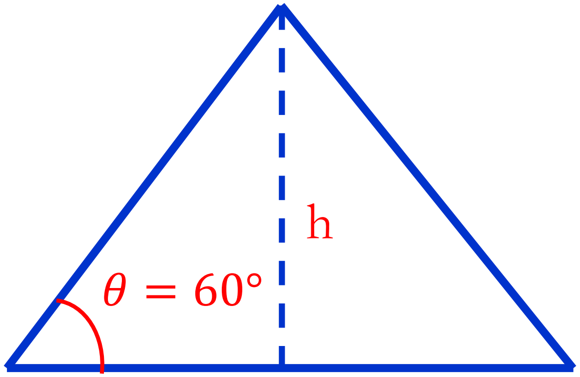


From figure , base$=\sqrt{2^2 + 2^2} = 2\sqrt{2}$

height(h)$=\sqrt{2^2 + 2^2} sin60° = 2\sqrt{2} sin60°$

So, loop area $S = 1/2 \times$ base $\times$ height

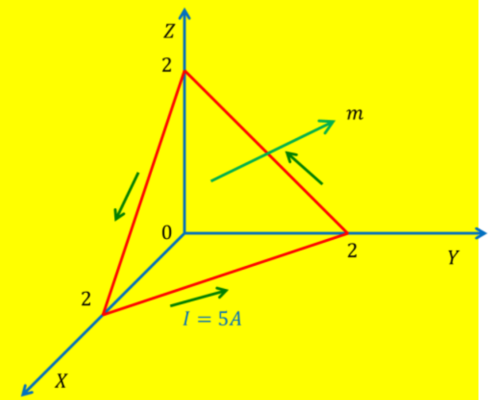
$$= 1/2 \times 2\sqrt{2} \times 2\sqrt{2} sin60°$$

$$= 4 \sin 60°$$

We define the plane surface by a function ,

$$f(x, y, z) = x + y + z - 2 = 0$$

Unit vector $\hat{n} = \pm \frac{\nabla f}{|\nabla f|} = \pm \frac{\hat{x} + \hat{y} + \hat{z}}{\sqrt{3}}$



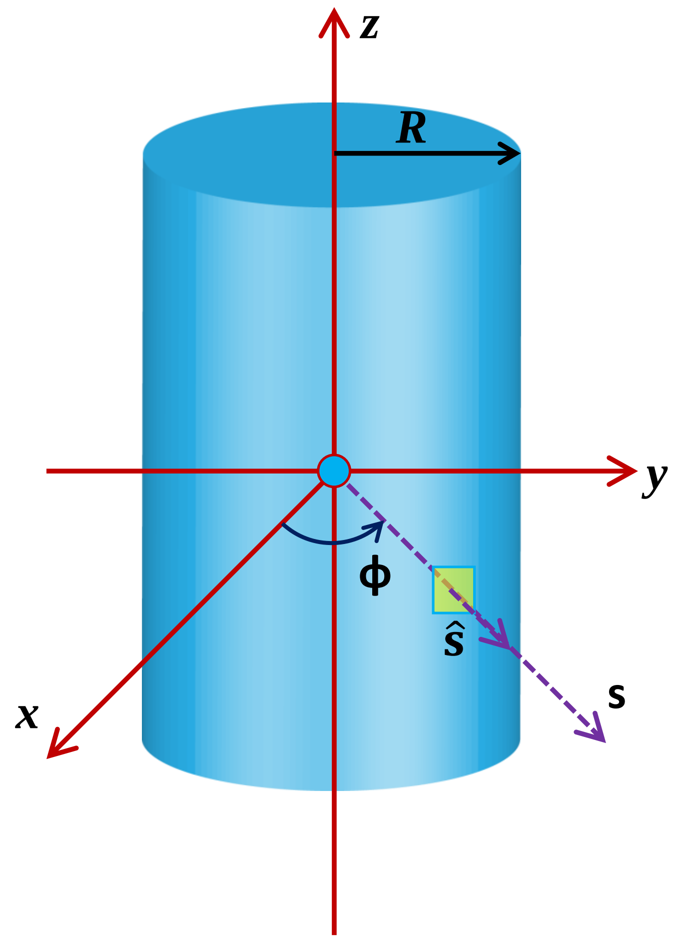
We choose the plus sign in view of the direction of the current in the loop (using the right-hand rule, m is directed as in Figure.

Magnetic moment of an electric circuit $\boldsymbol{m} = \mathrm{IS}\hat{n}$

$$= 5(4 sin60°)\left[\frac{\hat{x}+\hat{y}+\hat{z}}{\sqrt{3}}\right]$$

$$\mathbf{m} = \mathbf{10}(\hat{x} + \hat{y} + \hat{z}) \ \mathbf{A}.\boldsymbol{m}^2$$

Example: A long cylinder of radius **R** carries a magnetization $\vec{M} = ks^2\hat{\emptyset}$, where k is a constant, s is the distance from the axis, and $\hat{\emptyset}$ is the usual azimuthal unit vector (Figure 1). Find the magnetic field due to **M**, for points inside and outside the cylinder.



$$\vec{M} = ks^2\hat{\phi}$$

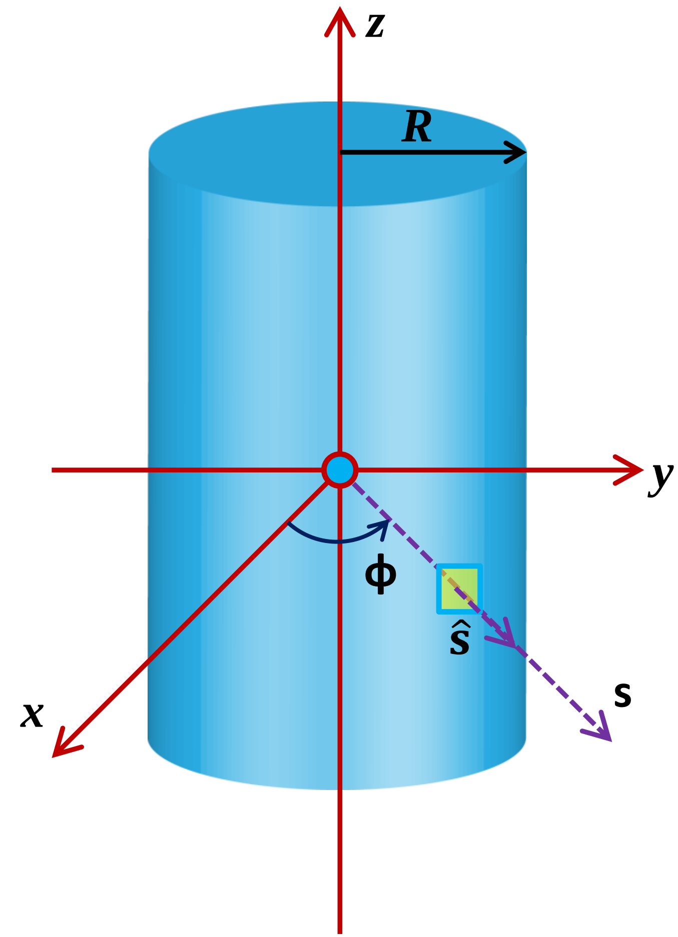
Using curl in cylindrical co-ordinates

$$J_b = \nabla \times \vec{M}$$

$$J_b = \frac{1}{s}\frac{\partial}{\partial s}(sks^2)\hat{z}$$

$$J_b = \frac{1}{s}(3ks^2)\hat{z}$$

$$J_b = 3ks\,\hat{z}$$



$$K_b = \vec{M} \times \hat{n}$$

$$K_b = ks^2\hat{\phi} \times \hat{s} \longrightarrow K_b = -kR^2\hat{z}$$

$$s = R\ (at\ surface)$$

So the bound current flows up the cylinder, and returns down the surface.

Total current should be ??

Let's check

$$\int J_b \, da = \int_0^R (3ks)(2\pi s) ds$$

$$\boxed{J_b = 3ks\,\hat{z}}$$

$$\int J_b \, da = 2\pi k R^3$$

$$\int K_b \, dl = (-kR^2)(2\pi R)$$

$$\boxed{K_b = -kR^2 \hat{z}}$$

$$\int K_b \, dl = -2\pi k R^3$$

$$\int J_b \, da + \int K_b \, dl = 2\pi k R^3 - 2\pi k R^3 = 0$$

$$\boxed{\oint \vec{B} \cdot dl = \mu_0 \, I_{enclosed}}$$ $\longrightarrow$ Now applying Ampere's Law

$$= \mu_0 \int J_b \, da$$

$$\boxed{J_b = 3ks \, \hat{z}}$$

$$= \mu_0 2\pi k s^3$$

$$\oint \vec{B} \cdot dl = B 2\pi s \;\; = \mu_0 2\pi k s^3$$

$$\boxed{B = \mu_0 k s^2 \widehat{\emptyset}}$$

Outside the cylinder

$$\oint \vec{B} \cdot dl = \mu_0 \, I_{enclosed}$$

$$I_{enclosed} = 0$$

$$B = 0$$

Problem 6.11

The Magnetic Field Inside Matter

Like the electric field, the actual microscopic magnetic field inside matter fluctuates wildly from point to point and instant to instant. When we speak of "the" magnetic field in matter. we mean the macroscopic field: the average over regions large enough to contain many atoms.

The magnetization M is "smoothed out" in the same sense.

We want the average of $B = B_{out} + B_{inside}$

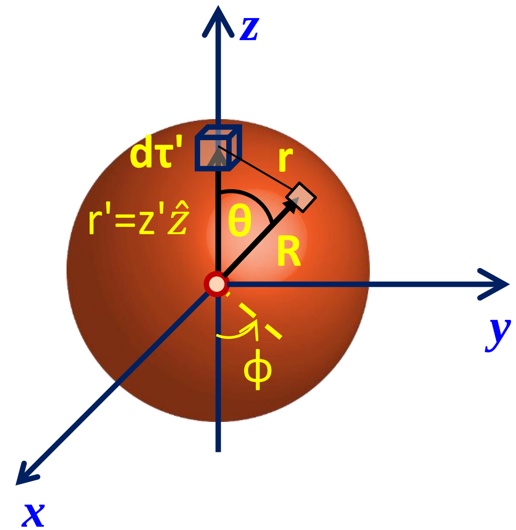
Here, B_{out} is the magnetic field due to the molecules outside a small sphere around Point P

And B_{Inside} is the magnetic field due to the molecules inside the sphere.

Proof: Average magnetic field over a sphere of radius R due to steady currents within the sphere

$$\boxed{B_{average} = \frac{1}{\frac{4}{3}\pi R^3}\int \vec{B}\, d\tau'}$$

$$\boxed{\vec{B} = \nabla \times \vec{A}}$$



$$B_{average} = \frac{3}{4\pi R^3} \int \nabla \times \vec{A}\, d\tau'$$

$$B_{average} = -\frac{3}{4\pi R^3} \oint \vec{A} \times \overrightarrow{da}$$

Griffith's Problem 1.60

$$\int (\nabla \times \mathbf{v})\, d\tau = -\int \mathbf{v} \times d\mathbf{a}$$

$$B_{average} = -\frac{3}{4\pi R^3} \oint \left[\frac{\mu_0}{4\pi} \int \frac{\overrightarrow{J(r')}}{r} d\tau' \right] \times \overrightarrow{da}$$

$$B_{average} = -\frac{3\mu_0}{(4\pi)^2 R^3} \int \overrightarrow{J(r')} \times \oint \left[\frac{1}{r} da \right] d\tau'$$

J depends only upon the source point r', not on the field point **r**

$$r = \sqrt{R^2 + z'^2 - 2\,R\,z'Cos\,\theta}$$

$$da = \; R^2\;\sin\theta\;d\theta\;d\emptyset\;\hat{r}$$

Since the z-component of $\hat{r}$ is cos θ

$$\oint \left[\frac{1}{r}da\right] = \hat{z}\int \frac{\cos\theta}{\sqrt{R^2 + z'^2 - 2\,R\,z'\cos\theta}} R^2\;\sin\theta\;d\theta\;d\emptyset$$

$$= 2\pi R^2\hat{z}\int_0^{\pi} \frac{\cos\theta\sin\theta}{\sqrt{R^2 + z'^2 - 2\,R\,z'\cos\theta}} d\theta$$

Let u = cos θ,
du = - sin θ dθ

$$= 2\pi R^2\hat{z}\int_{-1}^{+1} \frac{u}{\sqrt{R^2 + z'^2 - 2\,R\,z'u}} du$$

$$= 2\pi R^2 \hat{z} \left\{ \frac{-2\left[2(R^2 + {z'}^2 + 2\,R\,z'u)\right]}{3(2Rz')^2} \sqrt{R^2 + z'^2 - 2\,R\,z'u} \right\} \Bigg|_{-1}^{+1}$$

$$= \frac{2\pi R^2}{3(2Rz')^2} \hat{z} \left\{ \begin{array}{l} \left[(R^2 + {z'}^2 + 2\,R\,z')\right]\sqrt{R^2 + z'^2 - 2\,R\,z'} \\ -\left[(R^2 + {z'}^2 - 2\,R\,z')\right]\sqrt{R^2 + z'^2 + 2\,R\,z'} \end{array} \right\}$$

$$= \frac{2\pi}{3(z')^2} \hat{z} \left\{ \begin{array}{l} \left[(R^2 + {z'}^2 + 2\,R\,z')\right]|R - z'| \\ -\left[(R^2 + {z'}^2 - 2\,R\,z')\right]|R + z'| \end{array} \right\}$$

$$\oint \left[\frac{1}{r} d\mathrm{a}\right] = \frac{4\pi z'}{3} \hat{z} = \frac{4\pi}{3} \overrightarrow{r'} \qquad r' < \mathrm{R}$$

$$= \frac{4\pi R^3}{3(z')^2} \hat{z} = \frac{4\pi R^3}{3(r')^2} \overrightarrow{r'} \qquad r' > \mathrm{R}$$

$$\boxed{B_{average} = -\frac{3\mu_0}{(4\pi)^2 R^3}\int \overrightarrow{J(r')} \times \oint \left[\frac{1}{r} da\right] d\tau'}$$

For $r' < R$

$$B_{average} = -\frac{3\mu_0}{(4\pi)^2 R^3}\frac{4\pi}{3}\int \overrightarrow{J(r')} \times \vec{r'} d\tau'$$

$$B_{average} = -\frac{\mu_0}{4\pi R^3}\int \underbrace{\overrightarrow{J(r')} \times \vec{r'}}_{\text{Substituting}} d\tau'$$

We know that

$$\boxed{\mathbf{m} = \frac{1}{2}\int \vec{r'} \times \overrightarrow{J(r')}\, d\tau'}$$

Griffith's equation (5.91)

$$\boxed{B_{average} = \frac{\mu_0 2\mathbf{m}}{4\pi R^3}}$$

For $r' < R$

$$\boxed{B_{average} = \frac{\mu_0 \mathbf{m}}{2\pi R^3}}$$

Comparing this B_{avg} with the value obtained in Example problem 6.1 (previous Lecture)

Magnetic field inside the sphere $\longrightarrow$ $$\mathbf{B} = \frac{2}{3}\mu_0\mathbf{M}$$

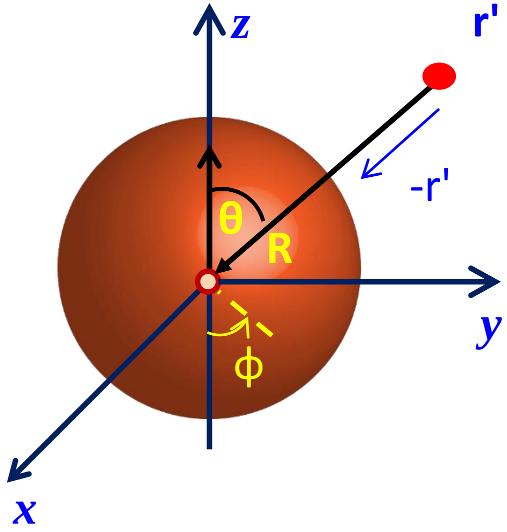
$$\mathbf{m} = \frac{4}{3}\pi R^3\mathbf{M} \Rightarrow \mathbf{M} = \frac{3\mathbf{m}}{4\pi R^3}$$

$$\boxed{\mathbf{B} = \frac{\mu_0\mathbf{m}}{2\pi R^3}} \Rightarrow \text{Same as } B_{avg} \Rightarrow \boxed{B_{average} = \frac{\mu_0\mathbf{m}}{2\pi R^3}}$$

For $r' > R$

$$B_{average} = -\frac{3\mu_0}{(4\pi)^2 R^3}\frac{4\pi}{3}R^3 \int \overrightarrow{J(r')} \times \frac{\overrightarrow{r'}}{r'^3} d\tau'$$

$$B_{average} = \frac{\mu_0}{4\pi}\int \frac{\overrightarrow{J(r')} \times \hat{r}}{r^3} d\tau'$$



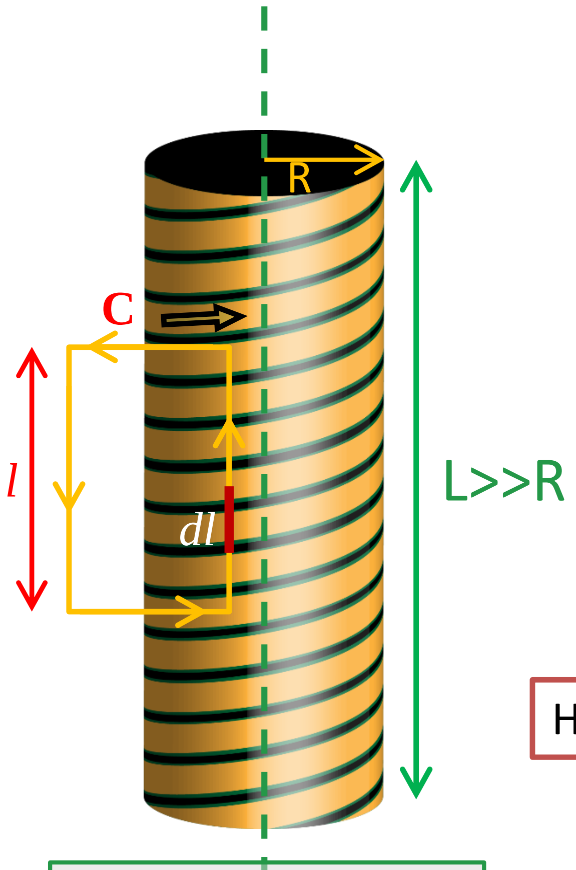
Now r goes from the source point to the center.

Thus

$$B_{average} = B_{Center}$$

Example: Comparison of Magnetic Field Strengths 'B' for a Long Solenoid With and Without a Magnetic Core (χ_m=1000)

Consider a long "ideal" solenoid of radius R and length L >> R has 'n' turns/meter, carrying a steady current $I = I\hat{\phi}$ (i.e. ideal solenoid – for simplicity's sake neglect pitch angle here). If solenoid has "soft" annealed iron magnetic core of magnetic permeability χ_m



Long Solenoid With Magnetic Core

$$\boxed{\mu = \mu_0(1 + \chi_m)}$$

$$\boxed{\mu = \mu_0(1 + 1000)}$$

$$\boxed{\mu = 1001\mu_0}$$

$$\overrightarrow{H_{inside}} = \frac{1}{\mu_0}\overrightarrow{B_{inside}} - \vec{M}$$

Here $\vec{M} = \chi_m H_{inside}$

$$\overrightarrow{H_{inside}} = \frac{1}{\mu}\overrightarrow{B_{inside}}$$

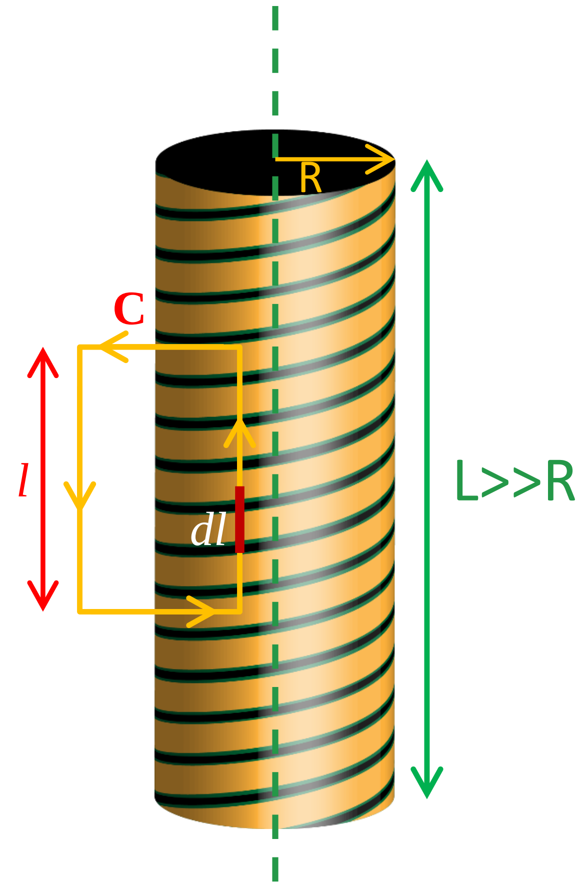
Long Solenoid Without Magnetic Core

Ampere's Circuital Law for H (integral form) with counter integral C as shown in the figure

$$\oint \vec{H} \cdot \vec{dl} = I_{f-enclosed}$$

The symmetry permits

$$\overrightarrow{H_{inside}} \parallel \hat{z}$$



Shrink contour C to ε above/below windings, then get:

$$H_{inside} l = nlI$$

$$\overrightarrow{H_{inside}}(r) = nI\hat{z}$$

Then

$$\overrightarrow{B_{inside}}(r) = \mu\overrightarrow{H_{inside}}(r)$$

$$\overrightarrow{B_{inside}}(r) = \mu nI\hat{z}$$

$$\boxed{\vec{M}(r) = \chi_m \overrightarrow{H_{inside}}}$$

$$\boxed{\vec{M}(r) = 1000\, nI\hat{z}}$$

Note that

$$\boxed{\overrightarrow{B(r)_{inside}} \, \Big\| \, \overrightarrow{H_{inside}} \, \Big\| \, \overrightarrow{M_{inside}} \, \| \, \hat{z}}$$

For long "ideal" solenoid with "soft" annealed iron core

Comparing these results to that for a long ideal solenoid without a magnetic core (for the same steady current *I*):

$$\boxed{\vec{H}_{inside}^{no\ core}(r) = \vec{H}_{inside}^{with\ core}(r) = nI\hat{z}}$$

$$\boxed{\vec{B}_{inside}^{no\ core}(r) = \mu_0 \vec{H}_{inside}^{no\ core}(r) = \mu_0 nI\hat{z}}$$

Thus

$$\frac{\left|\vec{B}_{inside}^{no\ core}(r)\right|}{\left|\vec{B}_{inside}^{with\ core}(r)\right|} = \frac{\mu_0\left|\vec{H}_{inside}^{no\ core}(r)\right|}{\mu\left|\vec{H}_{inside}^{with\ core}(r)\right|}$$

Since

$$\vec{H}_{inside}^{no\ core}(r) = \vec{H}_{inside}^{with\ core}(r) = nI\hat{z}$$

$$\frac{\left|\vec{B}_{inside}^{no\ core}(r)\right|}{\left|\vec{B}_{inside}^{with\ core}(r)\right|} = \frac{\mu_0\left|\vec{H}_{inside}^{no\ core}(r)\right|}{\mu\left|\vec{H}_{inside}^{with\ core}(r)\right|} = \frac{\mu_0}{\mu}$$

$$\boxed{\frac{\left|\vec{B}_{inside}^{no\ core}(r)\right|}{\left|\vec{B}_{inside}^{with\ core}(r)\right|} = \frac{1}{1001}}$$

$$\mu = 1001\mu_0$$

Another way to look at / view this: If one wants $\left|\vec{B}_{inside}^{with\ core}(r)\right| = \left|\vec{B}_{inside}^{no\ core}(r)\right|$, then one can reduce the current in the solenoid that has e.g. a "soft" annealed iron magnetic core by a factor of $\frac{\mu_0}{\mu} = \frac{1}{1001}$

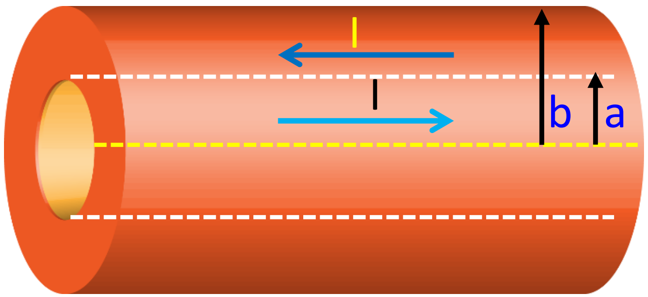
Example: A coaxial cable consists of two very long cylindrical tubes, separated by linear insulating material of magnetic susceptibility χ_m. A current I flows down the inner conductor and returns along the outer one; in each case the current distributes itself uniformly over the surface (Figure). Find the magnetic field in the region between the tubes. As a check, calculate the magnetization and the bound currents, and confirm that (together, of course, with the free currents) they generate the correct field.

$$\oint \vec{H} \cdot \vec{dl} = I_{f-enclosed}$$

$$\oint \vec{H} \cdot \vec{dl} = I$$

$$\mathrm{H} = \frac{I}{2\pi s}\hat{\phi}$$

$$\boxed{\vec{B} = \mu_0(1+\chi_m)\vec{H}}$$



Figure

$$\boxed{\vec{B} = \mu_0(1 + \chi_m)\frac{I}{2\pi s}\hat{\emptyset}}$$

$$\vec{M} = \chi_m \vec{H}$$

$$\vec{M} = \chi_m \frac{I}{2\pi s}\hat{\emptyset}$$

$$\vec{J_b} = \nabla \times \vec{M}$$

$$J_b = \frac{1}{s}\frac{\partial}{\partial s}\left(s\chi_m \frac{I}{2\pi s}\right)\hat{z}$$

Using curl in cylindrical co-ordinates

$$J_b = 0$$

$$K_b = \vec{M} \times \hat{n}$$

$$K_b = \chi_m \vec{H} \times \hat{n}$$

$$K_b = \chi_m \frac{I}{2\pi s} \hat{\phi} \times \hat{n}$$

$$\hat{\phi} \times \hat{s} = \hat{z}$$

$$K_b = \chi_m \frac{I}{2\pi a} \hat{z}$$

At **s = a**

$$K_b = -\chi_m \frac{I}{2\pi b} \hat{z}$$

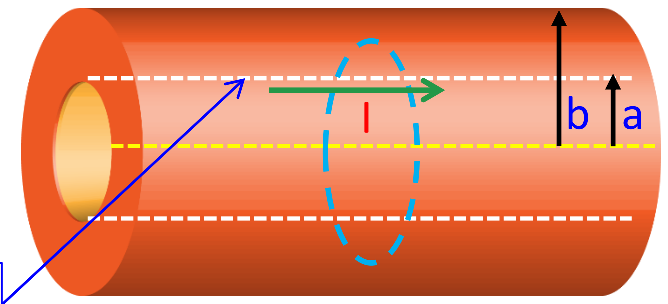
At **s = b** (-sign is due to current opposite direction)

Total enclosed current, for an Amperian loop between the cylinders:

$$I_{enclosed} = I + \chi_m \frac{I}{2\pi a} 2\pi a$$

$$\boxed{I_{enclosed} = I + \chi_m I}$$

Due to the inner surface



$$\oint \vec{B} \cdot \overrightarrow{dl} = \mu_0 I_{enclosed}$$

$$B\ 2\pi s = \mu_0 I_{enclosed} \longleftarrow \boxed{I_{enclosed} = I + \chi_m I}$$

$$B\ 2\pi s = \mu_0 (I + \chi_m I)$$

$$B = \frac{\mu_0 (1 + \chi_m) I}{2\pi s}$$

The direction of the magnetic field is : $\hat{\emptyset}$

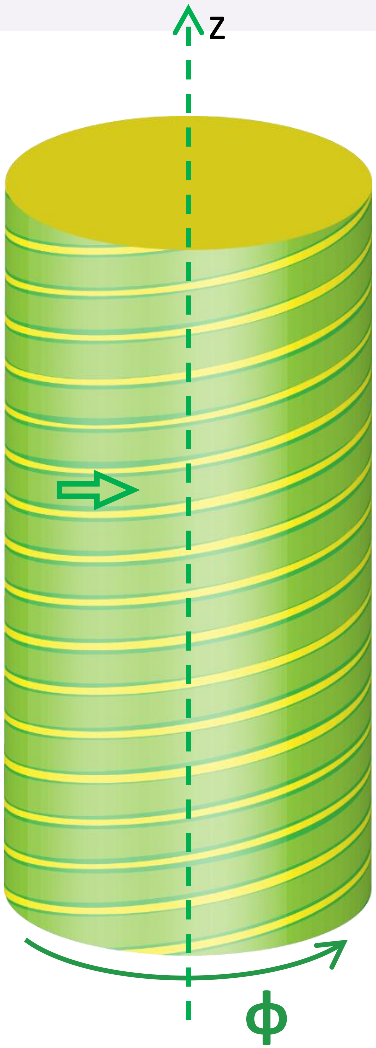
$$\boxed{B = \frac{\mu_0 (1 + \chi_m) I}{2\pi s} \hat{\emptyset}}$$

Same result obtained from both methods using Auxiliary field and Bound currents

$$\oint \vec{H} \cdot \overrightarrow{dl} = I_{f-enclosed}$$

Example: An infinite solenoid (n turns per unit length, current I) is filled with linear material of susceptibility χ_m. Find the magnetic field inside the solenoid.

Solution: Since B is due in part to bound currents (which we don't yet know), we cannot compute it directly. However, this is one of those symmetrical cases in which we can get H from the free current alone, using Ampere's law in the form of Eq.



$$\oint \vec{H} \cdot \vec{dl} = I_{f-enclosed}$$

$$H = nI\hat{z}$$

$$\vec{B} = \mu_0(1 + \chi_m)\vec{H}$$

$$\vec{B} = \mu_0(1 + \chi_m)nI\hat{z}$$

If the medium is paramagnetic, the field is slightly enhanced; if it's diamagnetic, the field is somewhat reduced. This reflects the fact that the bound surface current

$$\overrightarrow{K_b} = \vec{M} \times \hat{n}$$

$$\overrightarrow{K_b} = \chi_m(\vec{H} \times \hat{n})$$

$$\overrightarrow{K_b} = \chi_m\left(\vec{H} \times \hat{n}\right) = \chi_m \mathrm{nI}\widehat{\emptyset}$$

The same direction as I, in the former case ($\chi_m > 0$), and opposite in the latter ($\chi_m < 0$)

Example: A long copper rod of radius R carries a uniformly distributed (free) current I (Figure-1). Find **H** inside and outside the rod.

Solution: Copper is weakly diamagnetic, so the dipoles will line up opposite to the field.

This results in a bound current running antiparallel to I within the wire and parallel to I along the surface

However, we are not yet in a position to determine the magnitude of these bound currents

But in order to calculate **H** it is sufficient to realize that all the currents are longitudinal, so B, M, and therefore also H, are circumferential.

Applying the equation

$$\oint \vec{H} \cdot \vec{dl} = I_{f_{enclosed}}$$

For s<R

$$H(2\pi s) = I_{f_{enclosed}}$$

$$I_{f_{enclosed}} = I \frac{\pi s^2}{\pi R^2}$$

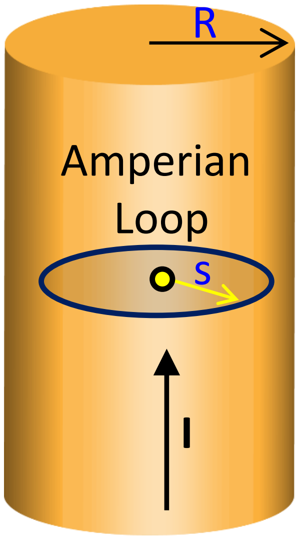


Figure-1

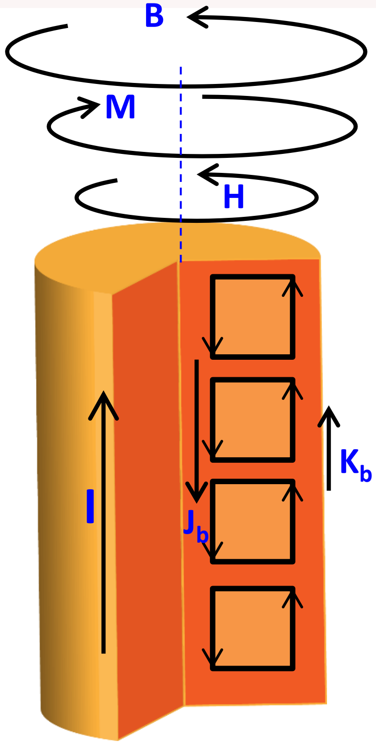


Figure-2 (Cross-sectional view)

$$H(2\pi s) = I\,\frac{\pi s^2}{\pi R^2}$$

Inside the wire

$$H = I\,\frac{s}{2\pi\;\;R^2}\,\hat{\emptyset}$$ For s ≤ R

Outside the wire

$$H = \frac{I}{2\pi s}\,\hat{\emptyset}$$ For s ≥ R

In the latter region (as always, in empty space) M = 0, so

$$\boxed{B = \mu_0 H = \frac{\mu_0 I}{2\pi s}\,\hat{\emptyset}}$$ Outside the wire

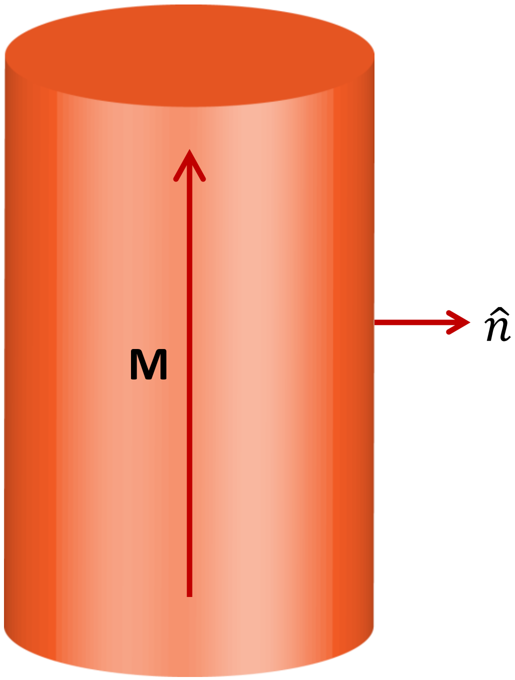
Inside the wire B cannot be determined at this stage, since we have no way of knowing M

Example: An infinitely long cylinder carries a uniform magnetization M parallel to its axis. Find the magnetic field (due to M) inside and outside the cylinder.

$$\mathrm{J_b} = \nabla \times \vec{\mathrm{M}} = 0$$

$$\mathrm{K_b} = \vec{\mathrm{M}} \times \hat{\mathrm{n}}$$

$$\mathrm{K_b} = \mathrm{M}\hat{\phi}$$



This is nothing but just a solenoid which we have solved previously in Lecture

So the field outside is zero

And field inside is

Upward direction

$$\mathrm{B} = \mu_0 \mathrm{K_b}$$

For Solenoid

$$\vec{B}(r \leq R) = \mu_0 n\, I$$

$$\vec{B}(r \geq R) = 0$$

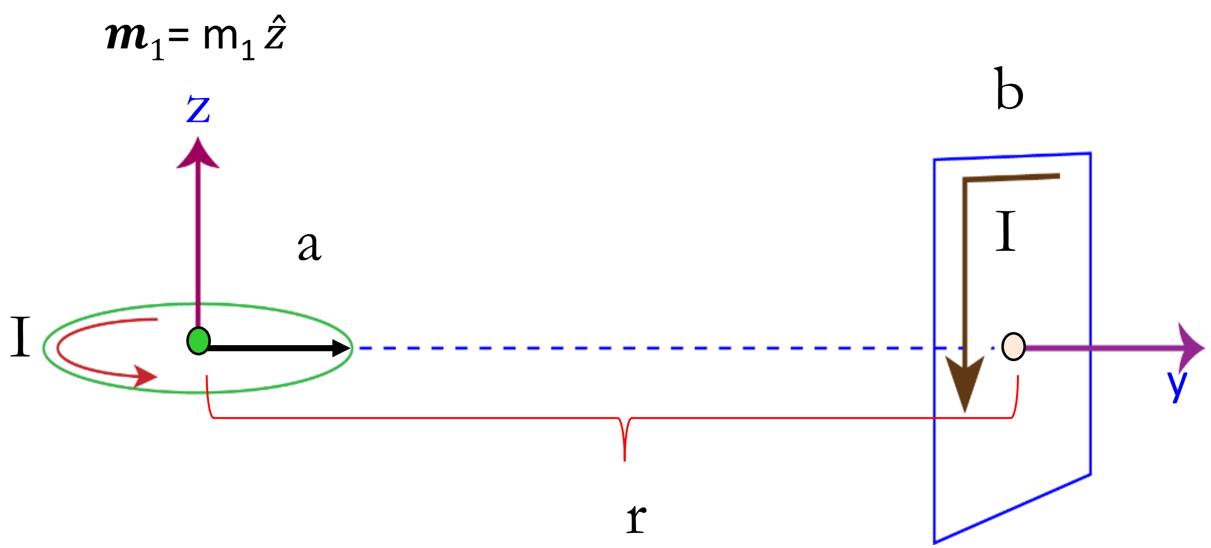
Example:

Calculate the Torque exerted on the square loop shown in the below figure, due to the circular loop (assume r is much larger than a or b). If the square loop is free to rotate, what will its equilibrium orientation be?

$\boldsymbol{m}_1 = m_1\,\hat{z}$

$\boldsymbol{m}_2 = m_2\,\hat{y}$

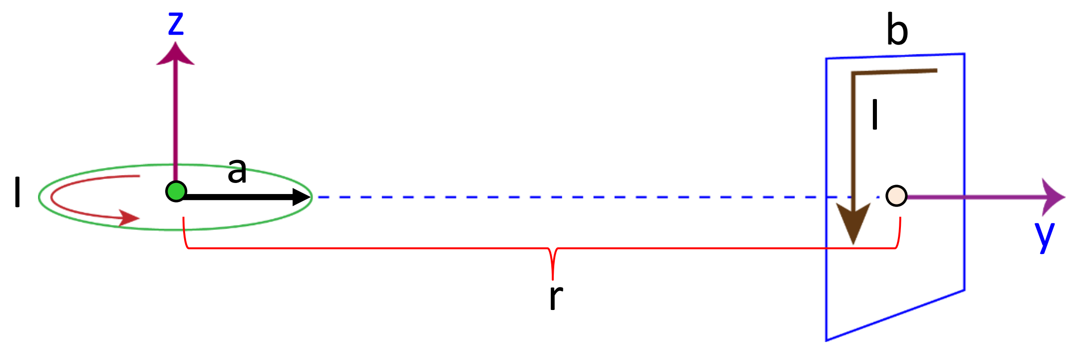
$\hat{r} = \hat{y}$



Solution 1:

Torque $\vec{N} = \vec{m} \times \vec{B}$

The magnetic field 'B' at a point 'r' due to a dipole having magnetic moment 'm'



$$\boldsymbol{B} = \frac{\mu_0}{4\pi}\frac{1}{r^3}[3(\boldsymbol{m}\cdot\hat{r})\hat{r} - \boldsymbol{m}]$$

Using this formula, the magnetic field 'B' at the center of square loop, due to the circular coil is

$\boldsymbol{m}_1 = \mathrm{m}_1\,\hat{z}$

$\boldsymbol{m}_2 = \mathrm{m}_2\,\hat{y}$

$\hat{r} = \hat{y}$

$$\boxed{\boldsymbol{B}_1 = \frac{\mu_0}{4\pi}\frac{1}{r^3}[3(\boldsymbol{m}_1\cdot\hat{r})\hat{r} - \boldsymbol{m}_1]}$$

$$\boldsymbol{B}_1 = \frac{\mu_0}{4\pi}\frac{1}{r^3}[3(\mathrm{m}_1\,\hat{z}\cdot\hat{y})\hat{y} - \mathrm{m}_1\,\hat{z}]$$

$$\boldsymbol{B_1} = \frac{\mu_0}{4\pi}\frac{1}{r^3}[-\mathrm{m}_1\hat{z}]$$

$\hat{z}\cdot\hat{y}=0$

$$\boldsymbol{B_1} = -\frac{\mu_0}{4\pi}\frac{1}{r^3}\mathrm{m}_1\hat{z}$$

Torque on the square loop

$$\vec{N} = \overrightarrow{m_2} \times \overrightarrow{B_1}$$

$$\vec{N} = \mathrm{m}_2\hat{y} \times -\frac{\mu_0}{4\pi}\frac{1}{r^3}\mathrm{m}_1\hat{z}$$

$\hat{y} \times \hat{z} = \hat{x}$

$$\boxed{\vec{N} = -\frac{\mu_0}{4\pi}\frac{1}{r^3}\mathrm{m}_1\mathrm{m}_2\hat{x}}$$

Here

$m_1 = \pi a^2 I$

$m_2 = b^2 I$

$$\vec{N} = -\frac{\mu_0}{4\pi}\frac{1}{r^3}\pi a^2 b^2 I^2 \hat{x}$$

$$\boxed{\vec{N} = -\frac{\mu_0}{4\pi}\frac{(abI)^2}{r^3}\hat{x}}$$

If the square loop is free to rotate, its equilibrium orientation will be in the direction of applied external magnetic field i.e. Magnetic field of the circular coil at the square loop.

$$\boldsymbol{B}_1 = \underbrace{-\frac{\mu_0}{4\pi}\frac{1}{r^3} m_1 \hat{z}}_{\text{Downward (negative z-direction)}}$$